Problem A. 越狱

Time Limit 1000 ms
Mem Limit 128000 kB

Description

There are n prison cells, each housing one prisoner. There are m different religions, and each prisoner follows exactly one of them. If two prisoners in adjacent cells share the same religion, a jailbreak might happen. Your task is to find out how many possible arrangements could lead to a jailbreak.

The answer should be given modulo $100,003$.

Input

The input consists of a single line with two integers: the number of religions m and the number of cells n.

Output

Output a single integer representing the number of possible jailbreak scenarios.

Sample 1

Input	Output
2 3	6

Hint

Explanation for Sample Input and Output 1

State ID	Cell 1	Cell 2	Cell 3
1	Religion 1	Religion 1	Religion 1
2	Religion 1	Religion 1	Religion 2

State ID	Cell 1	Cell 2	Cell 3
3	Religion 1	Religion 2	Religion 2
4	Religion 2	Religion 1	Religion 1
5	Religion 2	Religion 2	Religion 2
6	Religion 2	Religion 2	Religion 1

Data Constraints and Notes

For 100% dataset, it is guaranteed that $1 \le m \le 10^8$ and $1 \le n \le 10^{12}$.

Problem B. 青原樱

Time Limit 1000 ms
Mem Limit 512000 kB

Background

> Beneath the starry river, all is fireflies and dust,
> I walk alone amidst the crowd, while you remain innocent and kind.
> If our meeting were painted with colors,
> On the green plains, cherry blossoms bloom like a sea of pink.

—— Yin Lin, "Qingyuan Sakura" (Cover by Ren Yi Da Ren)

Description

Fusu is someone who loves listening to ancient-style music while writing math problems, so this one is actually a classic from the 53rd contest.

Fusu wants to recreate the breathtaking sight of cherry blossoms blooming on Qingyuan, so he has prepared many **distinct** cherry saplings to plant in a single row.

There are n spots in this row where cherry trees can be planted, and Fusu has m saplings ready. Since cherry blossoms need plenty of space on both sides when in full bloom, no two saplings can be planted right next to each other. In other words, between any two saplings, there must be at least one empty spot.

Planting under these rules isn't hard, but Fusu is curious: how many valid ways are there to plant all m saplings? A planting scheme is valid if and only if it meets the spacing rule described above. If we number the saplings from $1, 2, 3, \ldots, m$, two schemes are considered different if the chosen planting positions differ or if the sequence of saplings (from left to right) differs.

To keep the answer manageable, output the result modulo p.

Input

Each input file contains exactly one test case.

The test case consists of a single line with four integers representing $type, n, m, p$ respectively, where $type$ is a parameter to help you identify the test point type, explained in the data constraints.

Output

Output a single integer — the number of valid planting schemes modulo p.

Sample 1

Input	Output
`1 3 2 19260718`	`2`

Hint

Explanation for Sample Input/Output 1

There are 2 cherry saplings and 3 planting spots. If saplings are numbered $1, 2$ and spots are numbered $1, 2, 3$, then the two valid schemes look like this:

Spot	1	2	3
Scheme 1	Sapling 1	Empty	Sapling 2
Scheme 2	Sapling 2	Empty	Sapling 1

Data Scale and Notes

This problem uses bundled test points with 6 subtasks.

Subtask ID	$n \le$	$m \le$	$type =$	Special Property	Subtask Score
1	1	1	0	Special Property 1	5
2	20	20	1	Special Property 1	15
3	400	200	2	None	20
4	2000	2000	3	None	20
5	2000000	1000000	4	Special Property 2	20

Subtask ID	$n \le$	$m \le$	$type =$	Special Property	Subtask Score
6	2000000	1000000	5	None	20

Special Property 1: Guarantees that the **actual** number of valid schemes (before modulo) does not exceed 10^6.

Special Property 2: Guarantees that p is a prime number.

For data where 100% holds, it is guaranteed that:

- $1 \le n \le 2 \times 10^6$.
- $1 \le m \le 10^6$.
- $1 \le p \le 10^9$.
- $1 \le m \le \lceil \frac{n}{2} \rceil$.

Tips

- Use appropriate data types to avoid overflow during calculations.
- The parameter $type$ can help you quickly determine the subtask ID.

Problem C. 盒子与球

Time Limit 1000 ms
Mem Limit 128000 kB

Description

There are r different boxes and n different balls. These n balls need to be placed into r boxes without any empty boxes. Determine the number of different ways this can be done.

Two ways of placing the balls are considered different if there exists a ball that is placed in different boxes in the two ways.

Input

There is only one line with two integers representing n and r.

Output

Output a single integer representing the answer.

Sample 1

Input	Output
3 2	6

Hint

Explanation for Sample Input/Output 1

With two boxes (numbered $1, 2$) and three balls (numbered $1, 2, 3$), there are six possible arrangements as follows:

Box Number	Arrangement 1	Arrangement 2	Arrangement 3	Arrangement 4	Arrangement 5	Arrangement 6
Box 1	Ball 1	Ball 2	Ball 3	Ball 2, 3	Ball 1, 3	Ball 1, 2
Box 2	Ball 2, 3	Ball 1, 3	Ball 1, 2	Ball 1	Ball 2	Ball 3

Constraints

For 100% data points, ensure $0 \leq r \leq n \leq 10$, and the answer is less than 2^{31}.

Problem D. 栈

Time Limit 1000 ms
Mem Limit 128000 kB

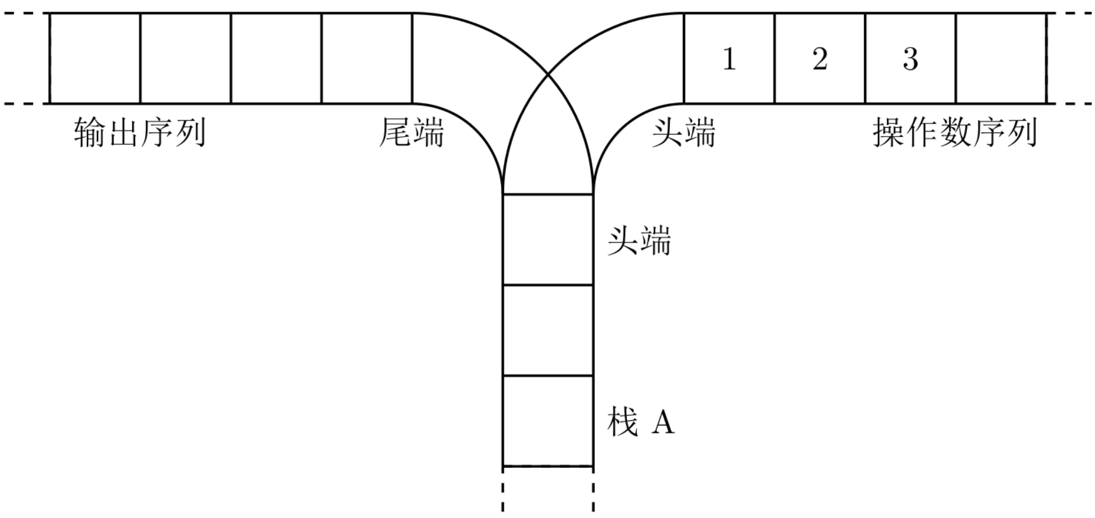
Background

A stack is a classic data structure in computer science. Simply put, a stack is a linear list that restricts insert and delete operations to one end.

There are two most important operations for a stack: pop (to remove an element from the top of the stack) and push (to add an element to the top of the stack).

The importance of stacks is self-evident, and any course on data structures will introduce stacks. During his review of the basic concepts of stacks, Ning Ning thought of a question that was not covered in the book, and he could not provide an answer himself, so he needs your help.

Description

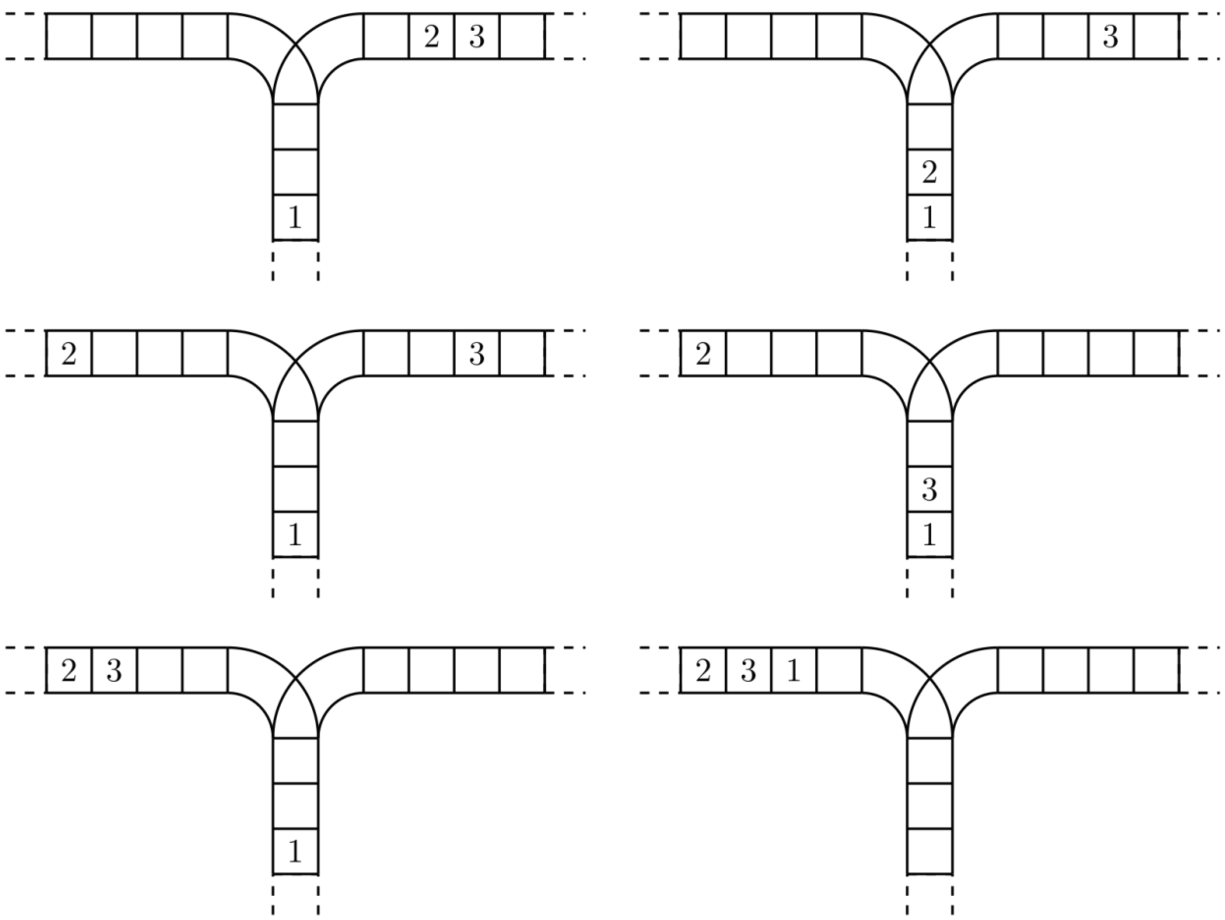


Ning Ning is considering the following problem: a sequence of operands, $1, 2, \ldots, n$ (illustrated for the case of 1 to 3), where the depth of stack A is greater than n.

Now, two operations can be performed:

1. Move a number from the head of the operand sequence to the head of the stack (corresponding to the push operation of the data structure stack).
2. Move a number from the head of the stack to the tail of the output sequence (corresponding to the pop operation of the data structure stack).

Using these two operations, a series of output sequences can be obtained from a sequence of operands. The following figure shows the process of generating the sequence `2 3 1` from `1 2 3`.



(The initial state is shown in the above figure.)

Your program will calculate and output the total number of possible output sequences that can be obtained from the operand sequence $1, 2, \ldots, n$ after performing operations.

Input

The input file contains only one integer n ($1 \leq n \leq 18$).

Output

The output file contains only one line, which is the total number of possible output sequences.

Sample 1

Input	Output
3	5

Hint

【Source of the Problem】

NOIP 2003 Popular Group Problem 3

Problem E. 消失之物

Time Limit 1000 ms

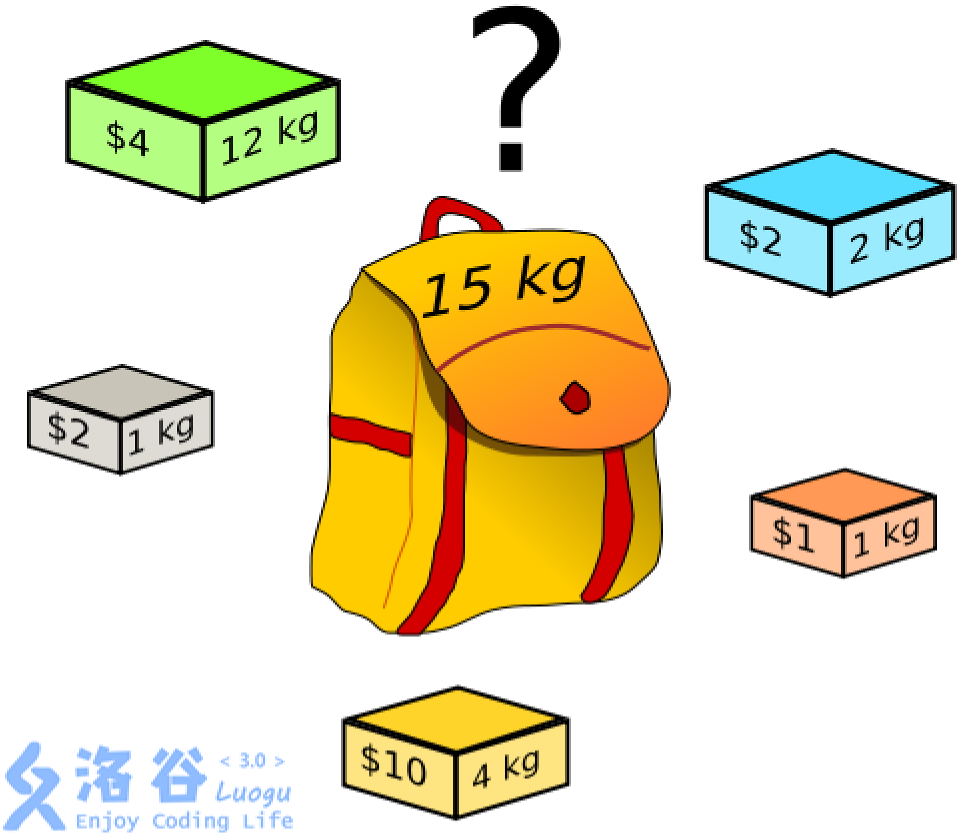
Mem Limit 262144 kB

Description

ftiasch has n items, each with a volume of $w_1, w_2, \dots, w_n$. Unfortunately, the i-rd item went missing.

"How many ways are there to fill a backpack of capacity x using the remaining $n-1$ items?" — a classic problem indeed.

She recorded the answer as $\text{cnt}(i, x)$ and wants to compute the entire $\text{cnt}(i, x)$ table for all $i \in [1, n]$ and $x \in [1, m]$ values.



Input

The first line contains two integers n, m, representing the number of items and the maximum capacity.
The second line contains n integers $w_1, w_2, \dots, w_n$, indicating the volume of each item.

Output

Print a $n \times m$ matrix showing the **last digit** of $\text{cnt}(i, x)$.

Sample 1

Input	Output
3 2 1 1 2	11 11 21

Hint

【Data Range】
For 100% data, $1 \leq n, m \leq 2000$, and $1 \leq w_i \leq 2000$.

【Sample Explanation】
If item 3 is missing, there's only one way to fill a backpack of capacity 2 — by choosing items 1 and 2.

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upd $2023.8.11$: Added five new hack test cases.

Problem F. 传球游戏

Time Limit 1000 ms
Mem Limit 128000 kB

Description

During physical education class, Xiaoman's teacher often plays games with the students. This time, the teacher organized a passing game.

The rules of the game are as follows: n students stand in a circle, and one of the students holds a ball. When the teacher blows the whistle, the passing begins. Each student can pass the ball to one of the two neighboring students (either left or right). When the teacher blows the whistle again, the passing stops, and the student who is holding the ball and has not passed it is the loser and must perform a show for everyone.

The clever Xiaoman posed an interesting question: how many different passing methods can be used such that the ball, starting from Xiaoman's hands, returns to Xiaoman after m passes? Two passing methods are considered different if and only if the sequence of students receiving the ball is different. For example, with three students 1, 2, and 3, assuming Xiaoman is 1, there are $1 \to 2 \to 3 \to 1$ and $1 \to 3 \to 2 \to 1$ ways for the ball to return to Xiaoman after 3 passes, totaling 2 methods.

Input

A single line containing two integers separated by a space $n, m (3 \le n \le 30, 1 \le m \le 30)$.

Output

1 integers representing the number of methods that meet the requirements of the problem.

Sample 1

Input	Output
3 3	2

Hint

Data Range and Conventions

- For the data of 40%, it satisfies: $3 \le n \le 30, 1 \le m \le 20$;
- For the data of 100%, it satisfies: $3 \le n \le 30, 1 \le m \le 30$.

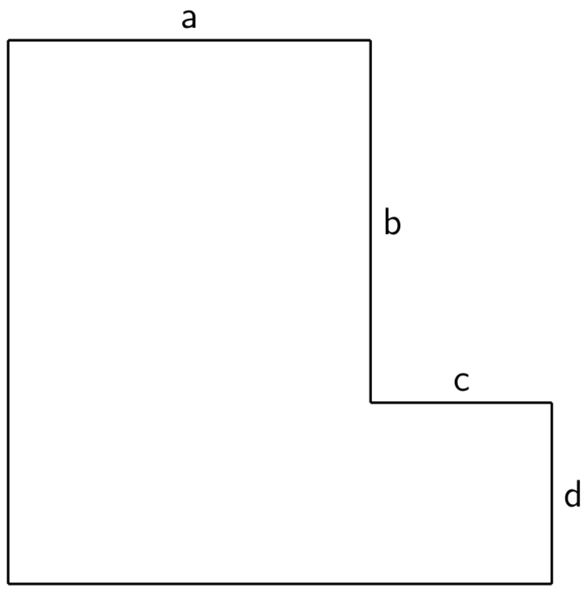
2008 Popular Group Problem 3

Problem G. 车的放置

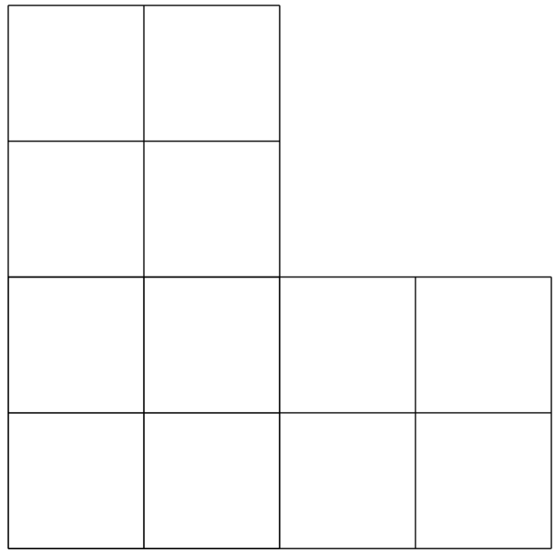
Time Limit 1000 ms
Mem Limit 128000 kB

Description

Imagine a grid-like chessboard where a, b, c, d represents the length of each side, or in other words, the number of squares along one edge:



When $a = b = c = d = 2$ is given, the board looks like this:



Your mission: place k rooks on this board so that none of them can attack each other. That means no two rooks share the same row or the same column. How many different ways can you do this?

Input

Just one line with five non-negative integers representing a, b, c, d and k respectively.

Output

Print a single integer — the number of valid arrangements $\bmod 10^5 + 3$ after applying the given operations.

Sample 1

Input	Output
2 2 2 2 2	38

Hint

Data Constraints and Notes

- Some test cases guarantee $b = 0$.
- Some test cases guarantee $a, b, c, d \le 4$.
- For data where 100% holds, it's guaranteed that $0 \le a, b, c, d, k \le 10^3$ and at least one valid arrangement exists.

Problem H. 计算系数

Time Limit 1000 ms
Mem Limit 128000 kB

Description

Given a polynomial $(by + ax)^k$, find the coefficient of the $x^n \times y^m$-th term after expanding the polynomial.

Input

The input consists of a single line containing 5 integers: a, b, k, n, m, separated by spaces.

Output

Output a single integer representing the coefficient you're looking for.

Since this coefficient can be huge, output the result modulo 10007.

Sample 1

Input	Output
1 1 3 1 2	3

Hint

【Data Constraints】

For 30% data, there is $0 \leq k \leq 10$.

For 50% data, there is $a = 1, b = 1$.

For 100% data, there is $0 \leq k \leq 1000, 0 \leq n, m \leq k, n + m = k, 0 \leq a, b \leq 10^6$.

This is Problem 1 from Day 2 of the NOIP 2011 Advanced Group.

Problem I. 组合数问题

Time Limit 1000 ms
Mem Limit 512000 kB

Background

NOIP2016 Advanced Group D2T1

Description

Combination number $\binom{n}{m}$ represents the number of ways to select m items from n items. For example, there are $(1, 2)$, $(1, 3)$, $(2, 3)$ ways to choose two items from $(1, 2, 3)$ three items. According to the definition of combination number, we can provide the general formula for calculating combination number $\binom{n}{m}$:

$$\binom{n}{m} = \frac{n!}{m!(n-m)!}$$

where $n! = 1 \times 2 \times \cdots \times n$; in particular, define $0! = 1$.

Xiao Cong wants to know given n, m and k, how many pairs of (i, j) satisfy $k \mid \binom{i}{j}$ for all $0 \le i \le n, 0 \le j \le \min(i, m)$.

Input

The first line contains two integers t, k, where t represents the total number of test data sets for this test point, and the meaning of k is explained in the problem description.

The next t lines each contain two integers n, m, where the meaning of n, m is explained in the problem description.

Output

There are t lines in total, each line containing an integer representing the number of pairs of (i, j) that satisfy $k \mid \binom{i}{j}$ among all $0 \le i \le n, 0 \le j \le \min(i, m)$.

Sample 1

Input	Output
`1 2` `3 3`	`1`

Sample 2

Input	Output

`2 5`
`4 5`
`6 7`
	`0`
	`7`

Hint

【Explanation for Sample 1】

Among all possible cases, there is only $\binom{2}{1} = 2$ case that is a multiple of 2.

【Subtasks】

测试点	n	m	k	t
1	≤ 3	≤ 3	$= 2$	$= 1$
2			$= 3$	$\le 10^4$
3	≤ 7	≤ 7	$= 4$	$= 1$
4			$= 5$	$\le 10^4$
5	≤ 10	≤ 10	$= 6$	$= 1$
6			$= 7$	$\le 10^4$
7	≤ 20	≤ 100	$= 8$	$= 1$
8			$= 9$	$\le 10^4$
9	≤ 25	≤ 2000	$= 10$	$= 1$
10			$= 11$	$\le 10^4$
11	≤ 60	≤ 20	$= 12$	$= 1$
12			$= 13$	$\le 10^4$
13	≤ 100	≤ 25	$= 14$	$= 1$
14			$= 15$	$\le 10^4$
15		≤ 60	$= 16$	$= 1$
16			$= 17$	$\le 10^4$
17	≤ 2000	≤ 100	$= 18$	$= 1$
18			$= 19$	$\le 10^4$
19		≤ 2000	$= 20$	$= 1$
20			$= 21$	$\le 10^4$

- For all test cases, ensure $0 \le n, m \le 2 \times 10^3$, $1 \le t \le 10^4$.

Problem J. Height All the Same

Time Limit 2000 ms
Mem Limit 524288 kB

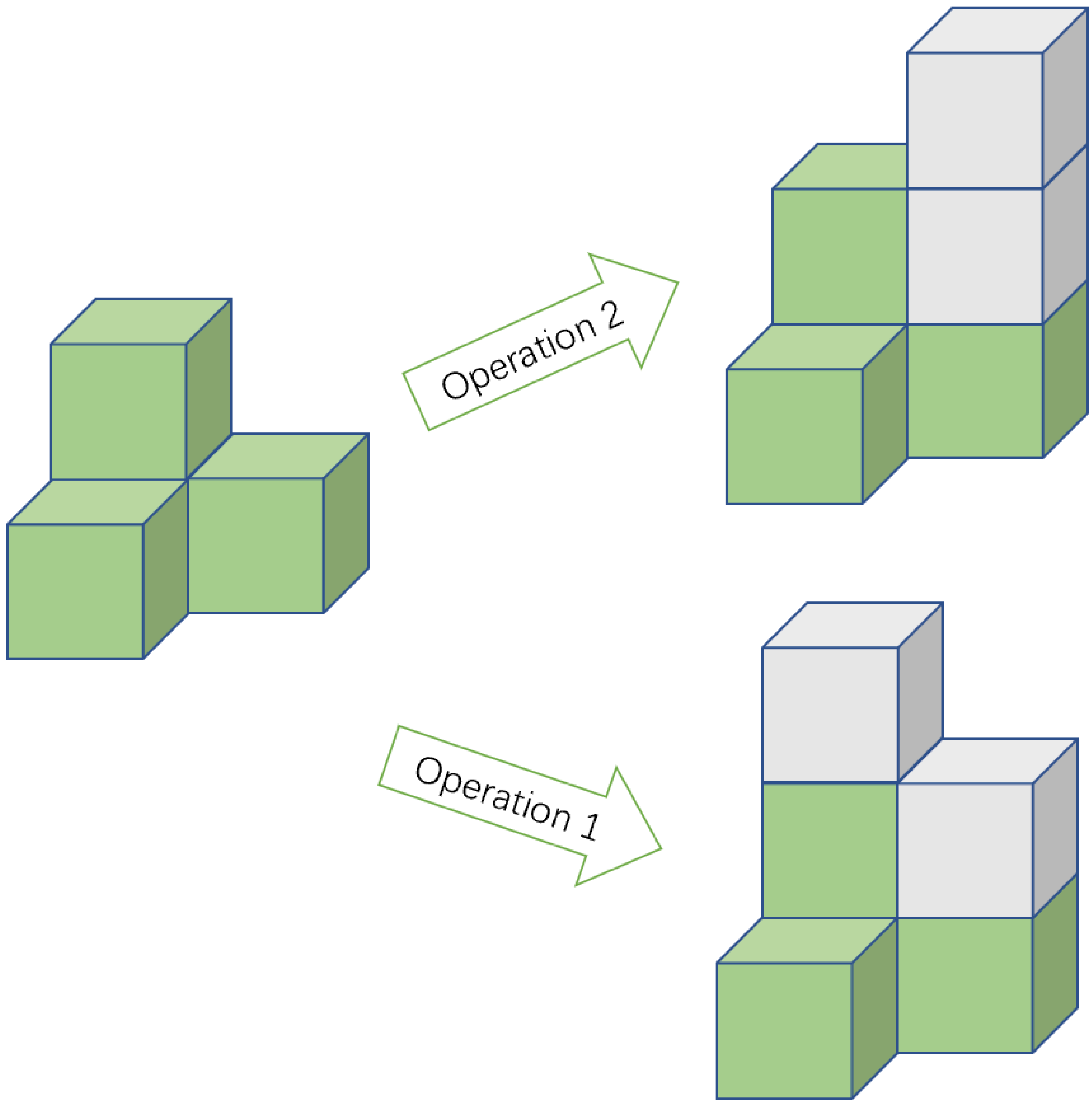
Alice has recently become hooked on a game called Sirtet.

In Sirtet, you start with an $n \times m$ grid. At the beginning, $a_{i,j}$ cubes are stacked in the cell (i, j). Two cells are considered adjacent if they share a common side. The player can perform the following moves:

- add one cube each to two **adjacent** cells;
- add two cubes to a single cell.

All cubes are identical in height.

Here's a quick visual of the game. The states on the right are reached by applying one of the moves above to the state on the left, with the newly added cubes shown in grey.



The player's goal is to **make all cells have the same height** — that is, every cell should contain the same number of cubes — by using the moves described above.

However, Alice has discovered that for some starting grids, no matter what strategy she tries, reaching this goal is impossible. So now she's curious: how many initial grids are there such that

- $L \le a_{i,j} \le R$ for every $1 \le i \le n$, $1 \le j \le m$;
- the player can reach the goal using the moves above.

Help Alice figure this out! Keep in mind the answer might be huge, so please output the result modulo $998,244,353$.

Input

The input consists of a single line with four integers: n, m, L, and R ($1 \le n, m, L, R \le 10^9$, $L \le R$, $n \cdot m \ge 2$).

Output

Print one integer — the answer modulo $998,244,353$.

Examples

Input	Output
2 2 1 1	1

Input	Output
1 2 1 2	2

Note

In the first example, the only initial grid that meets the conditions is $a_{1,1} = a_{2,1} = a_{1,2} = a_{2,2} = 1$. Therefore, the answer is 1.

In the second example, the initial grids that satisfy the conditions are $a_{1,1} = a_{1,2} = 1$ and $a_{1,1} = a_{1,2} = 2$. Hence, the answer is 2.

Problem K. 多边形

Time Limit 1000 ms
Mem Limit 524288 kB

Description

Little R loves playing with little sticks. Little R has n sticks, where the length of the i-th ($1 \leq i \leq n$) stick is a_i.

Little X wants Little R to pick some sticks from these n sticks and connect them end-to-end in any order to form a polygon. Little R doesn't know the condition for sticks to form a polygon, so Little X directly tells him: For m sticks with lengths $l_1, l_2, \ldots, l_m$, these m sticks can form a polygon if and only if the sum of all their lengths is **greater than** twice the length of the longest stick, that is, $m \geq 3$ and $\sum_{i=1}^{m} l_i > 2 \times \max_{i=1}^{m} l_i$.

Now that Little R knows the polygon condition, Little X throws a tougher challenge: How many ways are there to choose sticks so that the selected sticks can form a polygon? You need to help Little R find the number of such valid selections. Two selections are considered different if and only if their sets of stick indices differ, meaning there exists $1 \leq i \leq n$ such that one selection includes the i-th stick while the other does not. Since the answer could be huge, you only need to output the result modulo $998,244,353$.

Input

The first line contains a positive integer n, representing the number of sticks Little R has.

The second line contains n positive integers $a_1, a_2, \ldots, a_n$, representing the lengths of Little R's sticks.

Output

Output a single non-negative integer — the number of ways Little R can select sticks to form a polygon, modulo $998,244,353$.

Sample 1

Input	Output
5 1 2 3 4 5	9

Sample 2

Input	Output
5 2 2 3 8 10	6

Hint

【Explanation for Sample 1】

There are exactly 9 ways to select sticks that can form a polygon:

1. Choose stick number $2, 3, 4$, sum of lengths is $2+3+4=9$, max length is 4;
2. Choose stick number $2, 4, 5$, sum of lengths is $2+4+5=11$, max length is 5;
3. Choose stick number $3, 4, 5$, sum of lengths is $3+4+5=12$, max length is 5;
4. Choose stick number $1, 2, 3, 4$, sum of lengths is $1+2+3+4=10$, max length is 4;
5. Choose stick number $1, 2, 3, 5$, sum of lengths is $1+2+3+5=11$, max length is 5;
6. Choose stick number $1, 2, 4, 5$, sum of lengths is $1+2+4+5=12$, max length is 5;
7. Choose stick number $1, 3, 4, 5$, sum of lengths is $1+3+4+5=13$, max length is 5;
8. Choose stick number $2, 3, 4, 5$, sum of lengths is $2+3+4+5=14$, max length is 5;
9. Choose stick number $1, 2, 3, 4, 5$, sum of lengths is $1+2+3+4+5=15$, max length is 5.

【Explanation for Sample 2】

There are exactly 6 ways to select sticks that can form a polygon:

1. Choose stick number $1, 2, 3$, sum of lengths is $2+2+3=7$, max length is 3;
2. Choose stick number $3, 4, 5$, sum of lengths is $3+8+10=21$, max length is 10;
3. Choose stick number $1, 2, 4, 5$, sum of lengths is $2+2+8+10=22$, max length is 10;

4. Choose stick number $1, 3, 4, 5$, sum of lengths is $2 + 3 + 8 + 10 = 23$, max length is 10;
5. Choose stick number $2, 3, 4, 5$, sum of lengths is $2 + 3 + 8 + 10 = 23$, max length is 10;
6. Choose stick number $1, 2, 3, 4, 5$, sum of lengths is <v65/}, max length is 10.

【Sample 3】

See the files $polygon/polygon3.in$ and $polygon/polygon3.ans$ in the contestant directory.

This sample meets the constraints of test point $7 \sim 10$.

【Sample 4】

See the files $polygon/polygon4.in$ and $polygon/polygon4.ans$ in the contestant directory.

This sample meets the constraints of test point $11 \sim 14$.

【Subtasks】

For all test data, it is guaranteed that:

- $3 \le n \le 5,000$;
- For all $1 \le i \le n$, it holds that $1 \le a_i \le 5\,000$.

::cute-table{tuack}

Test Point ID	$n \le$	$\max_{i=1}^{n} a_i \le$
$1 \sim 3$	3	10
$4 \sim 6$	10	10^2
$7 \sim 10$	20	∧
$11 \sim 14$	500	∧
$15 \sim 17$	∧	1
$18 \sim 20$	$5\,000$	∧
$21 \sim 25$	∧	$5\,000$

Problem L. Help Yourself

Input File help.in
Output File help.out

Bessie has been given N segments ($1 \leq N \leq 10^5$) on a 1D number line. The ith segment contains all reals x such that $l_i \leq x \leq r_i$.

Define the **union** of a set of segments to be the set of all x that are contained within at least one segment. Define the **complexity** of a set of segments to be the number of connected regions represented in its union.

Bessie wants to compute the sum of the complexities over all 2^N subsets of the given set of N segments, modulo $10^9 + 7$.

Normally, your job is to help Bessie. But this time, you are Bessie, and there's no one to help you. Help yourself!

SCORING:

- Test cases 2-3 satisfy $N \leq 16$.
- Test cases 4-7 satisfy $N \leq 1000$.
- Test cases 8-12 satisfy no additional constraints.

INPUT FORMAT (file help.in):

The first line contains N.

Each of the next N lines contains two integers l_i and r_i. It is guaranteed that $l_i < r_i$ and all l_i, r_i are distinct integers in the range $1 \ldots 2N$.

OUTPUT FORMAT (file help.out):

Output the answer, modulo $10^9 + 7$.

Input	Output
3 1 6 2 3 4 5	8

The complexity of each nonempty subset is written below.

$$\{[1,6]\} \implies 1, \{[2,3]\} \implies 1, \{[4,5]\} \implies 1$$

$$\{[1,6],[2,3]\} \implies 1, \{[1,6],[4,5]\} \implies 1, \{[2,3],[4,5]\} \implies 2$$

$$\{[1,6],[2,3],[4,5]\} \implies 1$$

The answer is $1+1+1+1+1+2+1=8$.

Problem credits: Benjamin Qi

Problem M. 子集选取

Time Limit 1000 ms
Mem Limit 128000 kB

Description

Given a set n containing $S = \{1, 2, \cdots, n\}$ elements and an integer k, you need to pick several subsets S from $A_{i,j}$ ($A \subseteq S$ and arrange them into a triangle with side length k as shown below (so there are $\frac{1}{2}k(k+1)$ subsets in total):

$$\begin{array}{lllll} A_{1,1} & & & & \\ A_{2,1} & A_{2,2} & & & \\ A_{3,1} & A_{3,2} & A_{3,3} & & \\ \vdots & \vdots & \vdots & \ddots & \\ A_{k,1} & A_{k,2} & A_{k,3} & \cdots & A_{k,k} \end{array}$$

On top of that, JYY has some extra conditions for these chosen subsets: they must satisfy $A_{i,j} \subseteq A_{i,j-1}$ and $A_{i,j} \subseteq A_{i-1,j}$.
JYY wants to know how many different ways there are to select these subsets. Since the answer can be huge, JYY only cares about the result modulo 1,000,000,007.

Two selection schemes $A = \{A_{1,1}, A_{2,1}, \cdots, A_{k,k}\}$ and $B = \{B_{1,1}, B_{2,1}, \cdots, B_{k,k}\}$ are considered different if there exists at least one i, j such that $A_{i,j} \neq B_{i,j}$. In other words, A and B are different if they differ in at least one subset.

Input

The input consists of a single line with two integers: n and k.

Output

Output a single integer — the number of distinct selection schemes modulo $1,000,000,007$.

Sample 1

Input	Output
2 2	16

Hint

For the data where $100\%, 1 \leq n, k \leq 10^9$.

Problem N. 合唱队

Time Limit 1000 ms
Mem Limit 128000 kB

Description

In order to have a better performance at the upcoming party, as the leader of the AAA choir, Little A needs to arrange the choir members into a formation based on their heights. Assuming there are a total of n people in the choir, the height of the ith person is h_i meters ($1000 \le h_i \le 2000$), and it is known that the height of any two people is different. Assuming the final formation consists of A people standing in a row, to simplify the problem, Little A came up with the following way of arranging the queue: he first lets everyone stand in an initial formation in any order, and then from left to right, each person is inserted into the final formation according to the following principle:

- The first person is inserted directly into the empty current formation.
- For each person starting from the second person, if he is taller than the person in front of him (h larger), then he is inserted at the far right of the current formation. If he is shorter than the person in front of him (h smaller), then he is inserted at the far left of the current formation.

After n people are all inserted into the current formation, the final formation is obtained.

For example, if there are 6 people standing in an initial formation with heights
$1850, 1900, 1700, 1650, 1800, 1750,$
then Little A will obtain the final formation in the following steps:

- $1850.$
- $1850, 1900,$ because $1900 > 1850.$
- $1700, 1850, 1900,$ because $1700 < 1900.$
- $1650, 1700, 1850, 1900,$ because $1650 < 1700.$
- $1650, 1700, 1850, 1900, 1800,$ because $1800 > 1650.$
- $1750, 1650, 1700, 1850, 1900, 1800,$ because $1750 < 1800.$

Therefore, the final formation is $1750, 1650, 1700, 1850, 1900, 1800.$

Little A has an ideal formation in mind, and he wants to know how many initial formations can achieve the ideal formation.

Find the answer modulo 19650827.

Input

The first line contains an integer n.
The second line contains n integers, representing the ideal formation in Little A's mind.

Output

Output an integer in a single line, representing the value of the answer $\bmod 19650827$.

Sample 1

Input	Output
4 1701 1702 1703 1704	8

Hint

For data of 30%, $n \le 100$.
For data of 100%, $n \le 1000$, $1000 \le h_i \le 2000$.

Problem O. 排列计数

Time Limit 1000 ms
Mem Limit 128000 kB

Description

How many permutations of 1 through n are there such that the sequence has exactly m positions i where $a_i = i$ holds true?

The answer should be given modulo $10^9 + 7$.

Input

There are multiple test cases within a single test file.

The first line contains an integer T, representing the number of test cases.

The following T lines each describe a test case.

For each test case, a line contains two integers representing n and m respectively.

Output

Output T lines in total. For each test case, output a single integer representing the answer.

Sample 1

Input	Output
5 1 0 1 1 5 2 100 50 10000 5000	0 1 20 578028887 60695423

Hint

Data scale and constraints

There are 20 test points in total, evenly divided. The data scale for each test point is shown in the table below.

Test Point ID	$T =$	$n, m \le$	**Test Point ID**	$T =$	$n, m \le$
$1 \sim 3$	10^3	8	$10 \sim 12$	10^3	10^3
$4 \sim 6$	10^3	12	$13 \sim 14$	5×10^5	10^3
$7 \sim 9$	10^3	100	$15 \sim 20$	5×10^5	10^6

For all test points, it is guaranteed that $1 \le T \le 5 \times 10^5$, $1 \le n \le 10^6$, and $0 \le m \le 10^6$.

Problem P. 排队

Time Limit 1000 ms
Mem Limit 131072 kB

Description

At a certain middle school, there are n male students, m female students, and two teachers lining up for a health check. They form a single line, but with a twist: no two female students can stand side by side, and the two teachers can't be neighbors either. So, how many different ways can they line up? (Keep in mind: every individual is unique!)

Input

Just one line containing two non-negative integers n and m, separated by a space, representing the numbers as described above.

Output

A single non-negative integer showing the total number of valid arrangements. Heads up: the answer might be huge!

Sample 1

Input	Output
1 1	12

Hint

For the data 30%, $n \le 100$, $m \le 100$.

For the data 100%, $n \le 2000$, $m \le 2000$.

Problem Q. 树屋阶梯

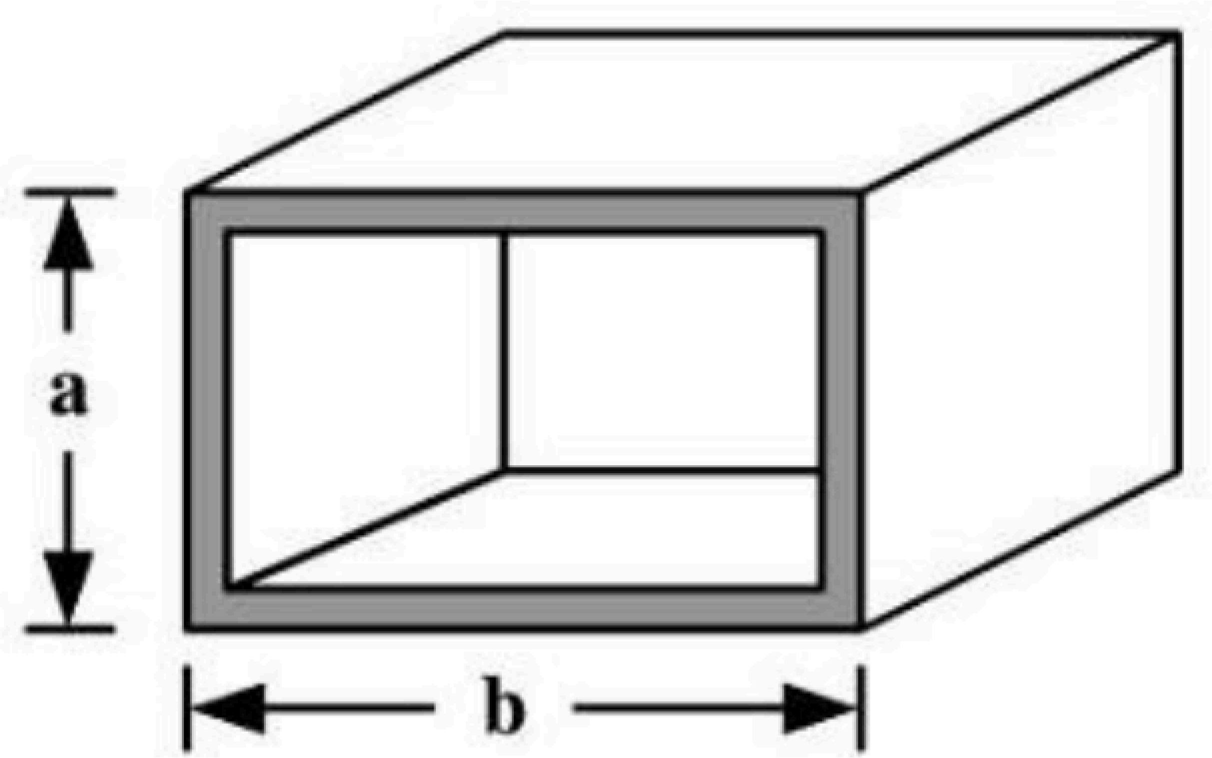
Time Limit 1000 ms
Mem Limit 128000 kB

Description

During summer vacation, Xiaolong enrolled in a simulated wilderness survival training course. On the first night of training, the instructor gave him a task: due to the slippery ground, the team members had to camp in a treehouse at a high altitude.

Xiaolong's assigned treehouse was built on a large tree with a height of $n+1$. To climb up, he found some hollow rectangular steel bars piled up nearby. Each steel bar had a fixed height of 1 and a positive integer width, and there were enough steel bars of each width.

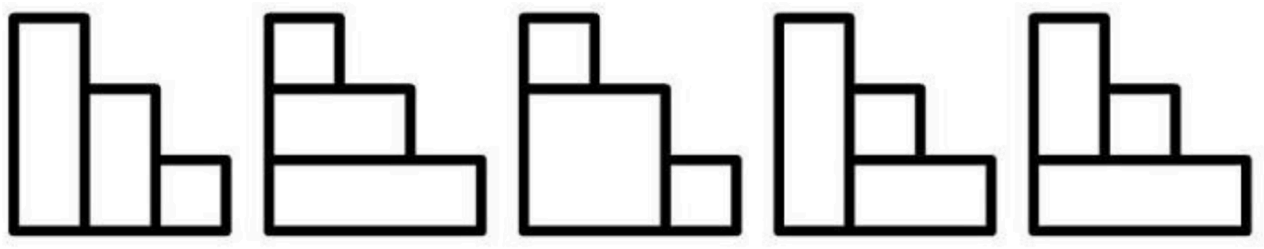
The instructor required each team member to build a staircase with a total height of n to reach the treehouse. The staircase consisted of n steps, each step having a height of 1 and a width of 1. For safety, **the hollow side of the steel bars must not face upwards.** (That is, the steel cannot be placed vertically).



Xiaolong can use any number of steel pieces to build each step by stacking them horizontally. A valid staircase must satisfy the following conditions:

1. Each step has a height of exactly 1 and a width of exactly 1.
2. The steel pieces are placed horizontally and cannot be rotated.
3. The staircase extends continuously from the ground to the treehouse.

For example, with $n = 3$, we have the following 5 Number of ways to build the stairs.



Given the height n of the stairs, calculate the number of different ways Xiaolong can build them. Since the result may be very large, high precision may be necessary.

Input

A positive integer n, representing the height of the stairs.

$1 \le n \le 500$.

Output

A positive integer, representing the number of valid ways to build the stairs.

Hint

- There exists a test case with 40%, where $n \le 20$.
- There exists a test case with 80%, where $n \le 300$.

Sample 1

Input	Output
3	5

Problem R. 有趣的数列

Time Limit 1000 ms
Mem Limit 128000 kB

Description

We call a sequence of length $2n$ "interesting" if and only if it meets the following three criteria:

- It is a permutation $\{a_n\}_{n=1}^{2n}$ of $2n$ integers taken from $1 \sim 2n$;
- All elements in odd positions satisfy $a_1 < a_3 < \cdots < a_{2n-1}$, and all elements in even positions satisfy $a_2 < a_4 < \cdots < a_{2n}$;
- For any two adjacent elements a_{2i-1} and a_{2i}, the condition $a_{2i-1} < a_{2i}$ holds.

Given n, your task is to find how many distinct interesting sequences of length $2n$ exist. Since the result could be huge, output the answer modulo p.

Input

A single line containing two positive integers n, p.

Output

Print a single integer representing the answer.

Sample 1

Input	Output
3 10	5

Hint

【Data Constraints】
For 50%, $1 \leq n \leq 1000$;
For 100%, $1 \leq n \leq 10^6$, $1 \leq p \leq 10^9$.

【Sample Explanation】

The corresponding 5 interesting sequences are

$(1, 2, 3, 4, 5, 6), (1, 2, 3, 5, 4, 6), (1, 3, 2, 4, 5, 6), (1, 3, 2, 5, 4, 6), (1, 4, 2, 5, 3, 6)$.

Problem S. Cave Paintings

Input File cave.in
Output File cave.out

Bessie has become an artist and is creating paintings of caves! Her current work in progress is a grid of height N such that each row of the grid contains exactly M squares ($1 \leq N, M \leq 1000$). Each square is either empty, filled with rock, or filled with water. Bessie has already painted the squares containing rock, including the entire border of the painting. She now wants to fill some empty squares with water such that if the painting were real, there would be no net motion of water. Define the height of a square in the i-th row from the top to be $N + 1 - i$. Bessie wants her painting to satisfy the following constraint:

Suppose that square a is filled with water. Then if there exists a path from a to square b using only empty or water squares that are not higher than a such that every two adjacent squares on the path share a side, then b is also filled with water.

Find the number of different paintings Bessie can make modulo $10^9 + 7$. Bessie may fill any number of empty squares with water, including none or all.

SCORING:

- Test cases 1-5 satisfy $N, M \leq 10$.
- Test cases 6-15 satisfy no additional constraints.

INPUT FORMAT (file cave.in):

The first line contains two space-separated integers N and M.

The next N lines of input each contain M characters. Each character is either '.' or '#', representing an empty square and a square filled with rock, respectively. The first and last row and first and last column only contain '#'.

OUTPUT FORMAT (file cave.out):

A single integer: the number of paintings satisfying the constraint modulo $10^9 + 7$.

Input	Output
4 9 ######### #...#...# #.#...#.# #########	9

If a square in the second row is filled with water, then all empty squares must be filled with water. Otherwise, assume that no such squares are filled with water. Then Bessie can choose to fill any subset of the three horizontally contiguous regions of empty squares in the third row. Thus, the number of paintings is equal to $1 + 2^3 = 9$.

Problem credits: Daniel Zhang

Problem T. Emiya 家今天的饭

Time Limit 1000 ms
Mem Limit 256000 kB

Description

Emiya is a high school student who is good at cooking. He has mastered n different **cooking methods** and uses m different **main ingredients** to cook. For the sake of convenience, we number the cooking methods from $1 \sim n$ and the main ingredients from $1 \sim m$.

Each dish made by Emiya will use **exactly one** cooking method and **exactly one** main ingredient. More specifically, Emiya will make $a_{i,j}$ different dishes using cooking methods i and main ingredients j ($1 \le i \le n, 1 \le j \le m$). This also means that Emiya will make a total of $\sum\limits_{i=1}^{n} \sum\limits_{j=1}^{m} a_{i,j}$ different dishes.

Today, Emiya is preparing a meal to entertain his good friends Yazid and Rin. However, the three of them have different requirements for the food pairing. Specifically, for a combination of k dishes:

- Emiya will not let anyone go hungry, so he will make **at least one dish**, namely $k \ge 1$
- Rin wants to taste dishes made with different cooking methods, so she requests that each dish has **a different cooking method**
- Yazid does not want to taste too many dishes made with the same main ingredient, so he requests that each **main ingredient** is used in at most **half** of the dishes (i.e., $\lfloor \frac{k}{2} \rfloor$ dishes)

Here, $\lfloor x \rfloor$ represents the floor function, indicating the largest integer not exceeding x.

These requirements do not stump Emiya, but he wants to know how many different combinations of dishes that meet the requirements exist. Two combinations are considered different if at least one dish appears in one combination but not in the other.

Emiya has found you, so he asks for your help in calculating the number of different compliant combinations of dishes modulo a prime number $998,244,353$.

Input

The first line contains two integers separated by a single space, n, m.

From the 2nd line to the $n+1$th line, each line contains m integers separated by single spaces. The $i+1$th line contains m integers in sequence.

Output

Only one line with a single integer, representing the result of the number of combinations modulo $998,244,353$.

Sample 1

Input	Output
2 3 1 0 1 0 1 1	3

Sample 2

Input	Output
3 3 1 2 3 4 5 0 6 0 0	190

Sample 3

Input	Output
5 5 1 0 0 1 1 0 1 0 1 0 1 1 1 1 0 1 0 1 0 1 0 1 1 0 1	742

Hint

【Explanation for Sample 1】

In this sample, for each group i, j, Emiya will make at most one dish, so we can describe a dish by providing the numbers of the cooking method and main ingredient.

The compliant combinations include:

- Making a dish using cooking method 1, main ingredient 1, and a dish using cooking method 2, main ingredient 2
- Making a dish using cooking method 1, main ingredient 1, and a dish using cooking method 2, main ingredient 3
- Making a dish using cooking method 1, main ingredient 3, and a dish using cooking method 2, main ingredient 2

Therefore, the output result is $3 \bmod 998,244,353 = 3$. It is important to note that all combinations consisting of only one dish are not compliant because the sole main ingredient appears in more than half of the dishes, which does not meet Yazid's requirement.

【Explanation for Sample 2】

Emiya must make at least 2 dishes.

The number of compliant combinations for making 2 dishes is 100.

The number of compliant combinations for making 3 dishes is 90.

Therefore, the total number of compliant combinations is 100 + 90 = 190.

【Data Constraints】

Test Case	$n=$	$m=$	$a_{i,j}<$	**Test Case**	$n=$	$m=$	$a_{i,j}<$
1	2	2	2	7	10	2	10^3
2	2	3	2	8	10	3	10^3
3	5	2	2	$9\sim 12$	40	2	10^3
4	5	3	2	$13\sim 16$	40	3	10^3
5	10	2	2	$17\sim 21$	40	500	10^3
6	10	3	2	$22\sim 25$	100	2×10^3	998244353

For all test cases, it is guaranteed that $1 \le n \le 100$, $1 \le m \le 2000$, $0 \le a_{i,j} < 998,244,353$.

Problem U. 排列计数

Time Limit 1000 ms
Mem Limit 131072 kB

Description

A permutation of a $1 \sim n$ is called Magic if and only if

$$\forall i \in [2, n], p_i > p_{\lfloor i/2 \rfloor}$$

Calculate how many permutations of $1 \sim n$ are Magic; the answer may be large, so output the value modulo m.

Input

One line containing two integers n, m, with the meanings as described above.

Output

The output file should contain only one integer, representing the number of Magic permutations of $1 \sim n$ modulo m.

Sample 1

Input	Output
20 23	16

Hint

【Data Range】
For the data of 100%, $1 \leq n \leq 10^6$, $1 \leq m \leq 10^9$, m is a prime number.

Problem V. 序列计数

Time Limit 3000 ms
Mem Limit 131072 kB

Description

Alice wants to create a sequence of length n where each number is a positive integer no greater than m. The sum of these n numbers must be divisible by p.

On top of that, Alice insists that at least one number in the sequence is a prime number.

She's curious—how many sequences fit her criteria?

Input

A single line with three numbers: n, m, p.

Output

Print one number—the count of sequences that satisfy Alice's conditions, modulo 20170408.

Sample 1

Input	Output
3 5 3	33

Hint

For 20% data, $1 \le n, m \le 100$.

For 50% data, $1 \le m \le 100$.

For 80% data, $1 \le m \le 10^6$.

For 100% data, $1 \le n \le 10^9, 1 \le m \le 2 \times 10^7, 1 \le p \le 100$.

Problem W. 硬币购物

Time Limit 1000 ms
Mem Limit 131072 kB

Description

There are 4 types of coins, with denominations of c_1, c_2, c_3, c_4.

Someone goes shopping n times. For each trip, they bring d_i coins of i different types, aiming to buy items worth s. Your task is to find out how many different ways they can pay each time.

Input

The first line contains five integers representing c_1, c_2, c_3, c_4, n.

Then, there are n lines, each with five integers describing a shopping trip, representing d_1, d_2, d_3, d_4, s.

Output

For each shopping trip, output a single integer on its own line indicating the number of payment methods.

Sample 1

Input	Output
1 2 5 10 2 3 2 3 1 10 1000 2 2 2 900	4 27

Hint

Data Scale and Constraints

- For data where 100%, it is guaranteed that $1 \le c_i, d_i, s \le 10^5$ and $1 \le n \le 1000$.

Problem X. 幸运数字

Time Limit 1000 ms
Mem Limit 128000 kB

Background

Sichuan NOI Provincial Selection 2010.

Description

In China, many folks consider 6 and 8 to be lucky digits! Our friend lxhgww thinks the same way, so he defines his own "lucky numbers" as those decimal numbers made up exclusively of the digits 6 and 8. For example, 68, 666, and 888 are all "lucky numbers"! But these "lucky numbers" are pretty rare — for instance, within the range $[1, 100]$, there are only 6 of them (6, 8, 66, 68, 86, 88). So, lxhgww came up with a new concept: "approximate lucky numbers." He says any multiple of a "lucky number" counts as an "approximate lucky number." Naturally, all "lucky numbers" themselves are also "approximate lucky numbers." For example, 12, 16, and 666 are all "approximate lucky numbers."

Now, lxhgww wants to find out how many "approximate lucky numbers" lie within a given closed interval $[a, b]$.

Input

The input consists of a single line containing 2 numbers: a and b.

Output

Output a single line with 1 number(s), representing the count of "approximate lucky numbers" within the closed interval $[a, b]$.

Sample 1

Input	Output
1 10	2

Hint

For 30% data, it is guaranteed that $1 \le a \le b \le 10^6$.

For 100% data, it is guaranteed that $1 \le a \le b \le 10^{10}$.

Problem Y. 分特产

Time Limit 1000 ms
Mem Limit 128000 kB

Description

JYY led a team to participate in several ACM/ICPC competitions and brought back a bunch of local specialties to share with the lab mates.

JYY is curious: if these goodies are divided among n students, how many unique ways can they be distributed? Of course, JYY wants to make sure no one feels left out, so every student must get at least one treat.

For example, JYY brought 2 bags of twisted dough sticks and 1 bags of buns, and wants to share them between two students, A and B. There are 4 different ways to split them:

A: twisted dough sticks, B: twisted dough sticks and buns

A: two twisted dough sticks, B: buns

A: buns, B: two twisted dough sticks

A: twisted dough sticks and buns, B: twisted dough sticks

Input

Input data:

The first line contains the number of students n and the number of specialty types m.

The second line has m integers, each representing the quantity of a particular specialty.

n, m does not exceed 1000, and the quantity of each specialty does not exceed 1000.

Output

Output a single line with the total number of different distribution methods.

Since the result can be huge, you only need to output the final answer modulo $\bmod 10^9 + 7$.

Sample 1

Input	Output
5 4 1 3 3 5	384835

Problem Z. Pólya 定理

Time Limit 1000 ms
Mem Limit 128000 kB

Description

Given a cycle with n vertices and n edges, and n colors to paint each vertex, how many **essentially different** coloring patterns are there? The answer should be given modulo $10^9 + 7$.

By "essentially different," we mean: two colorings are considered the same if one can be transformed into the other by rotation of the cycle.

Input

The first line contains an integer t, indicating there are t test cases.

Starting from the second line, there are t lines, each containing an integer n representing the parameters as described above.

Output

Output t lines, each line containing a number representing the count of distinct coloring schemes modulo $10^9 + 7$.

Sample 1

Input	Output
5 1 2 3 4 5	1 3 11 70 629

Hint

$$n \leq 10^9$$

$$t \le 10^3$$

Problem AA. Cards

Time Limit 1000 ms
Mem Limit 131072 kB

Description

Right now, Xiaochun is chilling with n cards on his desk. He decides to color each card, and luckily, he has 3 colors at his disposal: red, blue, and green. Curious, he asks Sun how many ways there are to color the cards, and Sun quickly spits out the answer.

Then Xiaochun ups the ante: he wants exactly S_r red cards, S_b blue cards, and S_g green cards. Again, he asks Sun for the number of possible colorings, and after a moment's thought, Sun nails the correct count. Finally, Xiaochun invents m different shuffling methods and wonders how many distinct colorings exist when considering these shuffles. Two colorings are considered the same if one can be transformed into the other by applying any combination of these shuffles (each shuffle can be used multiple times).

Sun finds this a bit tricky and passes the challenge to you. Since the answer might be huge, you only need to output the result modulo P. It's guaranteed that P is a prime number.

Input

The first line contains 5 integers representing: S_r, S_b, S_g, m, P ($m \leq 60, m+1 < p < 100$). Among them, the mentioned n equals $S_r + S_b + S_g$, that is, $n = S_r + S_b + S_g$.

Then follow m lines, each describing a shuffle method. Each line has n integers $X_1 X_2 ... X_n$ separated by spaces, guaranteed to be a permutation of numbers from 1 to n. This means that applying this shuffle moves the card originally at position X_i to position i. The input guarantees that any number of shuffles can be replaced by a combination of these m shuffles, and for each shuffle, there exists another shuffle that can undo it.

Additionally, for 100% data, it holds that $\max\{S_r, S_b, S_g\} \leq 20$.

Output

The number of distinct colorings modulo P.

Sample 1

Input	Output

`1 1 1 2 7`
`2 3 1`
`3 1 2`
	`2`

Hint

There are 2 essentially different colorings: `RGB` and `RBG`. Applying the shuffle `231` once transforms them into `GBR` and `BGR`, and applying the shuffle `312` once transforms them into `BRG` and `GRB`.

Problem AB. 项链

Time Limit 3000 ms
Mem Limit 128000 kB

Background

Necklaces have been one of the earliest forms of personal adornment. Beyond just looking good, some necklaces carry special meanings—like the cross chains for Catholics or prayer beads for Buddhists.

Throughout history, people have crafted necklaces in all sorts of styles and designs to beautify both the human body and the environment, catering to different skin tones, ethnicities, and tastes. When it comes to materials, necklaces on the market are typically made from gold, silver, gems, and so on.

Pearl necklaces, made by stringing drilled pearls together, are not only decorative but natural pearl necklaces are also believed to have some nurturing benefits.

Description

Lately, Mingming has become obsessed with a special kind of necklace. It's pretty much like any other pearl necklace, except the beads are unique—they're carved from Taishan stone shaped as regular triangular prisms.

Each triangular prism's side is a square, and each side is engraved with a number. For Mingming to be happy, the necklace must meet these conditions:

1. The necklace consists of n beads.
2. Every number x on each bead must satisfy $0 < x \leq a$, and the greatest common divisor (GCD) of the three numbers on each bead must be exactly 1.
3. Adjacent beads must be different. Two beads are considered the same if one can be turned or flipped to match the other.
4. Two necklaces are considered identical if one can be rotated to become the other.

Mingming is curious—given n and a, how many distinct necklaces can be made? Since the answer could be huge, output it modulo $10^9 + 7$.

Input

Multiple test cases.

The first line contains an integer T, the number of test cases.
Then T lines follow, each with two integers n and a.

Output

For each test case, output a single integer representing the number of distinct necklaces possible.

Sample 1

Input	Output
1 2 2	3

Hint

There are exactly three types of beads that meet the conditions: `[1,1,1]`, `[1,1,2]`, and `[1,2,2]`.

Possible valid sequences include: `[1,2]`, `[1,3]`, and `[2,3]`.

For the data in 100%, it is guaranteed that $1 \le T \le 10$, $2 \le n \le 10^{14}$, and $1 \le a \le 10^7$.

Problem AC. 集合划分计数

Time Limit 1000 ms
Mem Limit 128000 kB

Description

Given a set with n elements, divide it into any number of non-empty subsets. Your task is to find out how many different ways there are to do this.
Keep in mind, the order of these subsets doesn't matter — so $\{\{1,2\},\{3\}\}$ and $\{\{3\},\{2,1\}\}$ count as the same arrangement.

Since the answer can get huge, output it modulo 998244353.

Input

The first line contains a positive integer T, representing the number of test cases.
Then follow T lines, each containing a single positive integer n.

Output

Print T lines, each line showing the answer for the corresponding test case.

Sample 1

Input	Output
5 2 3 7 9 233	2 5 877 21147 53753544

Hint

【Data Constraints】
$T = 1000, 1 \le n \le 10^5$.

【Sample Explanation】

For $n = 3$, there are five ways:

$\{\{1,2,3\}\}, \{\{1,2\},\{3\}\}, \{\{1\},\{2,3\}\}, \{\{1\},\{2\},\{3\}\}, \{\{1,3\},\{2\}\}$.

This problem has only one test point. If you get x test cases right, you'll earn $\lfloor x/(T/100) \rfloor$ points.

Even if you can't solve all the data, please output T integers.

~~Don't blame me for TLE, blame your big constant factors!~~

Problem AD. 拯救世界

Time Limit 500 ms
Mem Limit 131072 kB

Background

In the year 2000 AD, according to the prophecy of Nostradamus, the world is about to end!!!

Description

To save the world, little a and uim decided to summon the legendary gods kkksc03 and lzn. Ancient texts say that summoning any of these gods requires arranging a special formation using the five elemental divine stones: Metal, Wood, Water, Fire, and Earth. The ancient scrolls describe the requirements as follows:

How to summon the god kkksc03:

- The number of Metal stones must be a multiple of 6;
- At most 9 Wood stones can be used;
- At most 5 Water stones can be used;
- The number of Fire stones must be a multiple of 4;
- At most 7 Earth stones can be used.

How to summon the god lzn:

- The number of Metal stones must be a multiple of 2;
- At most 1 Wood stones can be used;
- The number of Water stones must be a multiple of 8;
- The number of Fire stones must be a multiple of 10;
- At most 3 Earth stones can be used.

Now, it's the year 1999 AD, 12 month, 31 day. Little a and uim started searching from 00:00:00 and kept going until 23:00:00, but still couldn't find the divine stones.

However, when they got back home, they found some strange stuff in their cellar. Checking the ancient texts, oh boy, why didn't they come here sooner? They discovered some Chaos Stones that can be struck to decay into the five elemental stones. So they hammered away and finally produced n stones, completing two formations by 23:59:59.

Yet, although kkksc03 and lzn appeared, they lacked enough energy to unleash their powers. Only by arranging all possible formations using exactly n stones can they fully charge the gods. Now little a and uim are stumped and have reached out to you to help calculate how many different formations can be made.

Input

Input a positive integer n.

Output

Output the number of different formations that can be arranged using exactly n Chaos Stones.

Sample 1

Input	Output
2	15

Hint

Data Range and Constraints

For all data, $10^{99999} \leq n < 10^{100000}$.

Tips

Since it's already $23{:}59{:}59$, you only have 0.5 seconds. (Little a and uim need $0.5\mathtt{s}$ seconds to arrange all formations).

Problem AE. 拯救世界2

Time Limit 1000 ms
Mem Limit 128000 kB

Background

The first three problems were too easy, huh? Let's take a look at this final crappy one.

Description

After 12 years of laying low, the apocalypse is back again (everyone: what kind of logic is this...).

This time, little a and uim have prepared everything perfectly and successfully summoned the legendary kkksc03 and lzn. However, kkksc03 and lzn told them the apocalypse is just too powerful this time—they can't save the world. Only the Creator God JOHNKRAM can.

But the Creator God JOHNKRAM can't be summoned unless the entire universe is converted into energy $E = mc^2$. According to the summoning law casually derived by the great C_SUNSHINE, you need at least one ten-trillionth of the summonee's energy to call them forth (everyone: what kind of law is this...).

Of course, there's another way: find the gene sequence of the Creator God JOHNKRAM. Normal people's genes are made up of A, C, G, and T, but JOHNKRAM isn't normal (he's a chubby dude), so his gene sequence is different. Besides these four, there are eight special gene types: Qian, Dui, Li, Zhen, Xun, Kan, Gen, and Kun. Among them, Qian, Kan, Gen, and Zhen are Yang and must appear an odd number of times; Kun, Dui, Li, and Xun are Yin and must appear an even number of times.

Right now, we only know the Creator God JOHNKRAM's gene sequence length is n. Everything else is a mystery. Little a and uim want to know the maximum number of tries they'll need to summon JOHNKRAM. This number could be huge, so output the answer modulo 10^9 (C_SUNSHINE's advice: stay away from gossip, stay away from chubbiness).

Input

The input consists of multiple test cases, each on a separate line containing a number n. The input ends with 0.

Output

For each test case, output a single integer representing the answer modulo 10^9.

Sample 1

Input	Output
3 10 20 6 0	0 225116160 53238784 7680

Hint

【Data Range】
For 10% data: $1 \le n < 25$, no more than 10 test cases;
For 50% data: $1 \le n < 2^{31}$, no more than 10^3 test cases;
For 100% data: $1 \le n < 2^{63}$, no more than 2×10^5 test cases.

【Sample Explanation】
For the first test case:
Only 3 positions, no valid sequences, so the answer is 0.

【Notes】

Attachment: Chat log (pure nonsense)

JOHNKRAM 8:50:33

Hey hey, can the Pit God Contest 2 start now?

C_SUNSHINE 8:50:34

[Auto-reply] Yep!

JOHNKRAM 8:51:12

I'm planning to increase the last problem's data from 50 to 2^{63}.

C_SUNSHINE 8:51:12

[Auto-reply] Yep!

JOHNKRAM 8:51:45

You agree?

C_SUNSHINE 8:51:46

[Auto-reply] Yep!

C_SUNSHINE 11:58:50

Are you crazy?!

JOHNKRAM 11:58:52

[Auto-reply] Sorry, I'm busy right now. Will get back to you later. No more reminders.

Problem AF. Partitions

Time Limit 2000 ms
Mem Limit 262144 kB

You are given a set of n elements indexed from 1 to n. The weight of i-th element is w_i. The weight of some subset of a given set is denoted as $W(S) = |S| \cdot \sum_{i \in S} w_i$. The weight of some partition R of a given set into k subsets is $W(R) = \sum_{S \in R} W(S)$ (recall that a partition of a given set is a set of its subsets such that every element of the given set belongs to exactly one subset in partition).

Calculate the sum of weights of all partitions of a given set into exactly k **non-empty** subsets, and print it modulo $10^9 + 7$. Two partitions are considered different iff there exist two elements x and y such that they belong to the same set in one of the partitions, and to different sets in another partition.

Input

The first line contains two integers n and k ($1 \le k \le n \le 2 \cdot 10^5$) — the number of elements and the number of subsets in each partition, respectively.

The second line contains n integers w_i ($1 \le w_i \le 10^9$)— weights of elements of the set.

Output

Print one integer — the sum of weights of all partitions of a given set into k **non-empty** subsets, taken modulo $10^9 + 7$.

Examples

Input	Output
4 2 2 3 2 3	160

Input	Output
5 2 1 2 3 4 5	645

Note

Possible partitions in the first sample:

1. $\{\{1,2,3\},\{4\}\}$, $W(R)=3\cdot(w_1+w_2+w_3)+1\cdot w_4=24$;
2. $\{\{1,2,4\},\{3\}\}$, $W(R)=26$;
3. $\{\{1,3,4\},\{2\}\}$, $W(R)=24$;
4. $\{\{1,2\},\{3,4\}\}$, $W(R)=2\cdot(w_1+w_2)+2\cdot(w_3+w_4)=20$;
5. $\{\{1,3\},\{2,4\}\}$, $W(R)=20$;
6. $\{\{1,4\},\{2,3\}\}$, $W(R)=20$;
7. $\{\{1\},\{2,3,4\}\}$, $W(R)=26$;

Possible partitions in the second sample:

1. $\{\{1,2,3,4\},\{5\}\}$, $W(R)=45$;
2. $\{\{1,2,3,5\},\{4\}\}$, $W(R)=48$;
3. $\{\{1,2,4,5\},\{3\}\}$, $W(R)=51$;
4. $\{\{1,3,4,5\},\{2\}\}$, $W(R)=54$;
5. $\{\{2,3,4,5\},\{1\}\}$, $W(R)=57$;
6. $\{\{1,2,3\},\{4,5\}\}$, $W(R)=36$;
7. $\{\{1,2,4\},\{3,5\}\}$, $W(R)=37$;
8. $\{\{1,2,5\},\{3,4\}\}$, $W(R)=38$;
9. $\{\{1,3,4\},\{2,5\}\}$, $W(R)=38$;
10. $\{\{1,3,5\},\{2,4\}\}$, $W(R)=39$;
11. $\{\{1,4,5\},\{2,3\}\}$, $W(R)=40$;
12. $\{\{2,3,4\},\{1,5\}\}$, $W(R)=39$;
13. $\{\{2,3,5\},\{1,4\}\}$, $W(R)=40$;
14. $\{\{2,4,5\},\{1,3\}\}$, $W(R)=41$;
15. $\{\{3,4,5\},\{1,2\}\}$, $W(R)=42$.

Problem AG. 卡农

Time Limit 1000 ms
Mem Limit 131072 kB

Description

It is well known that Canon is a technique for composing polyphonic music. Xiaoyu was inspired while listening to Canon music and invented a new set of rules for music composition.

He divided the sound into n scales and divided the music into several segments. Each segment of the music is a harmony composed of 1 to n scales, meaning that several scales are selected from n scales to be played simultaneously.

To emphasize the difference from Canon, he stipulated that the sets of scales contained in any two segments must be different. Also, to maintain the regularity of the music, he further specified that each scale is played an even number of times in a piece of music.

Now, the question is: Xiaoyu wants to know how many different types of music can be created containing m segments.

Two pieces of music a and b are of the same type if and only if rearranging the a segments can result in b. For example: if a is $\{\{1,2\},\{2,3\}\}$, and b is $\{\{2,3\},\{1,2\}\}$, then a and b are of the same type.

The answer should be taken modulo 10^8+7.

Input

Only one line with two positive integers n, m

Output

Output a single line with an integer representing the answer.

Sample 1

Input	Output
2 3	1

Hint

【Data Range】
For data of 20%, $1 \le n, m \le 5$;
For data of 50%, $1 \le n, m \le 3000$;
For data of 100%, $1 \le n, m \le 10^6$.

【Example Explanation】
The music is $\{\{1\},\{2\},\{1,2\}\}$

Problem AH. Periodni

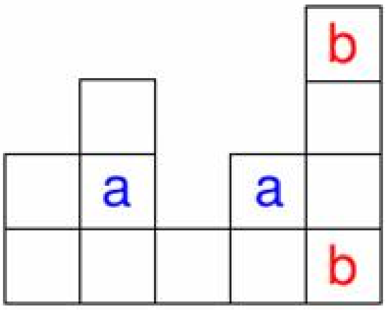
Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

English	Vietnamese

Luka is bored in chemistry class so he is staring at a large periodic table of chemical elements hanging from a wall above the blackboard. To kill time, Luka decided to make his own table completely different from the one in the classroom.

His table consists of N columns, each with some height, aligned at the bottom (see example below). After he draws the table he needs to fill it with elements. He first decided to enter the noble gases of which there are K. Luka must put them in the table so that no two noble gases are close to each other.

Two squares in the table are close to each other if they are in the same column or row, and all squares between them exist. In the example below, the 'a' squares are not close, but the 'b' squares are.



Write a program that, given N, K and the heights of the N columns, calculates the total number of ways for Luka to place the noble gases into the table. This number can be large, so output it modulo 1 000 000 007.

Input

The first line contains the integers N and K separated by a space ($1 \le N \le 500$, $1 \le K \le 500$), the number of columns in Luka's table and the number of noble gases.

The next line contains N positive integers, separated by spaces. These are heights of the columns from left to right. The heights will be at most 1 000 000.

Output

Output the number of ways for Luka to fill his table with noble gases, modulo 1 000 000 007.

Example

Input	Output
5 2 2 3 1 2 4	43

Problem AI. 落忆枫音

Time Limit 1000 ms
Mem Limit 128000 kB

Background

"Hengyi, do you believe in the existence of souls?" Guo Hengyi and Yao Fengxi were strolling down the streets of Maple Sound Village. As they watched the crimson maple leaves fluttering in the air, Fengxi suddenly asked this question.

"I guess I do. Otherwise, what are we? Just lumps of flesh? If there were no souls... we wouldn't even be able to see your sister again." Hengyi gave a somewhat quirky answer. Fengxi smiled softly. "Have you ever really looked closely at a maple leaf?" With that, Fengxi reached out and caught a falling maple leaf.

"Actually, every maple leaf has a soul. See all those veins on the leaf? I heard that some special spots on the leaf are like acupoints on a human body. The veins connect these acupoints. The soul of the maple tree flows through the base of each leaf, travels along these veins, slowly permeating the acupoints and saturating the whole leaf. That's why the veins only flow one way—souls can't flow backward." Hengyi nodded, half understanding. Fengxi continued.

"Because of the soul, every maple leaf is unique. And because of the soul, each leaf resembles its parent maple tree, even the veins form a tree-like pattern. But if you look closely, you'll find other very thin veins outside the main vein tree. Although these veins aren't part of the tree, their directions also follow the soul's flow—there are no loops that would cause the soul to flow backward." Hengyi suddenly seemed to have an idea. "So these veins could replace the existing ones and appear on the vein tree?" Fengxi closed her eyes.

"Exactly. The vein tree isn't unique. Even tiny differences can lead to vastly different vein trees, even within the same leaf. Just like our story, the ending isn't set in stone. Changing one small choice can completely twist the storyline."

"That's deep..." Hengyi stared at the red maple leaf, lost in thought. Fengxi went on.

"And that's not all. Veins don't last forever, nor do they disappear forever. Whether veins on the vein tree or the tiny ones outside, they can break and vanish, or reconnect again. Everything is in eternal flux, including the bonds between people. Maybe one day, our ties will be severed like veins, and we will become 'passersby of Maple Sound Village.' Maybe it's all inevitable, decided by the soul of the maple tree..."

Tears glistened in Fengxi's eyes. Hengyi looked at her and pulled her into his arms.

"Don't think like that, Fengxi. Even if veins break, new vein trees can form and reconnect to the maple tree's roots. Our bond still exists, just taking a longer path. No matter what, I won't leave you. Because you are the true love I've searched for my whole life!"

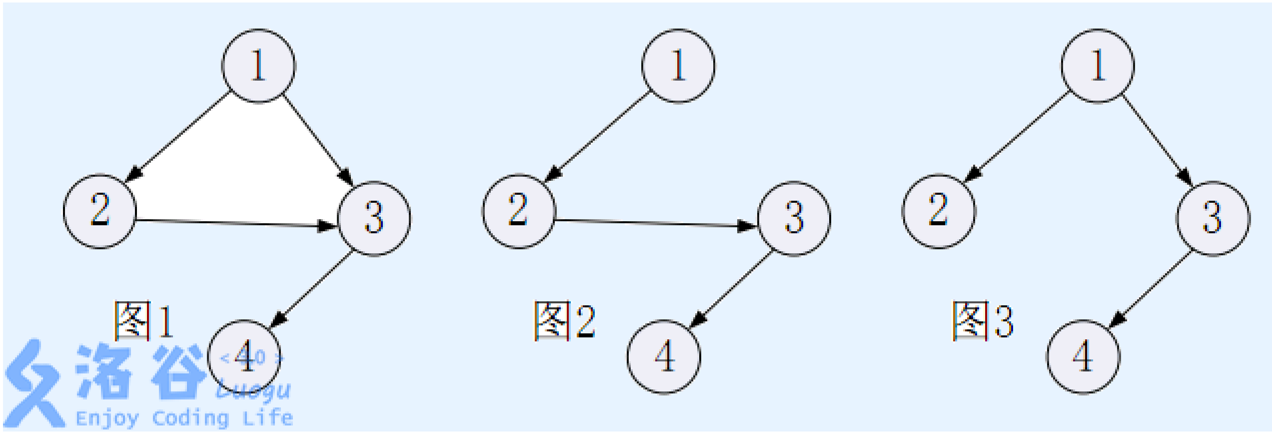
Their eyes met. Fengxi smiled happily and buried her head in Hengyi's embrace. From the distant maple forest on the mountain came the sound of the maples.

Description

Let's imagine there are n acupoints on a maple leaf, each labeled with an ID $1 \sim n$. Several directed veins connect these acupoints.

The acupoints and veins form a Directed Acyclic Graph (DAG)—called the **vein graph** (for example, see graph 1). The acupoint with ID 1 has no veins leading into it, meaning acupoint 1 only has outgoing veins. From the story, we know this DAG contains a tree-shaped subgraph rooted at acupoint 1 that includes all n acupoints—this is called the **vein tree** (for example, the trees shown in graphs 2 and 3 are subgraphs of the vein graph in 1).

Note that there might be multiple possible vein tree configurations in the vein graph. For example, graphs 2 and 3 show two different vein trees of the vein graph in 1.



Formally, a vein tree rooted at acupoint r consists of all n acupoints and $(n-1)$ veins, contains no cycles, and no vein connects an acupoint to itself. Moreover, for every acupoint s on the leaf, there exists a unique vein path within the vein tree starting from acupoint r and reaching acupoint s.

Now, we add a new vein to the vein graph, different from any existing vein (note: veins connecting the same two acupoints but in opposite directions are considered different veins—for example, a vein from acupoint 3 to 4 is different from one from 4 to 3; thus, in graph 1, you cannot add a vein from 3 to 4, but you can add one from 4 to 3). This new vein **can even connect an acupoint to itself** (for example, in graph 1, you can add a vein from 4 to 4).

After adding this new vein, the updated vein graph may contain cycles. Your task is to find the number of vein tree configurations rooted at acupoint 1 in this new vein graph.

Since the number of configurations can be huge, output the count modulo $(10^9 + 7)$.

Input

The first line contains four integers n, m, x, y, representing the number of acupoints on the maple leaf, the number of veins, and the new vein's start and end acupoints x and y.

The next m lines each contain two integers separated by a space, representing a vein.

The i-th line contains two integers u_i and v_i, indicating that the i-th vein goes from acupoint u_i to acupoint v_i.

Output

Output a single line containing the number of vein tree configurations rooted at acupoint x after adding the vein from acupoint y to acupoint 1, modulo $(10^9 + 7)$.

Sample 1

Input	Output
4 4 4 3 1 2 1 3 2 4 3 2	3

Hint

For all test data, it is guaranteed that:

- $1 \le n \le 100000$;
- $n-1 \le m \le \min(200000, n(n-1)/2)$;
- $1 \le x, y, u_i, v_i \le n$.

Fixed by Starrykiller.

Problem AJ. Let us count 1 2 3

Time Limit	1000 ms
Mem Limit	1572864 kB
Code Length Limit	50000 B
OS	Linux

Given two integer n, p. 4 kinds of query is needed to solve:

1. Counting the number of labeled unrooted trees with n nodes.
2. Counting the number of labeled rooted trees with n nodes.
3. Counting the number of unlabeled rooted trees with n nodes.
4. Counting the number of unlabeled unrooted trees with n nodes.

Calculate the answer modulo p.

Input

Each line contains three integers k, n, p. k denotes which kind of query this case is.

For Kind 1 or Kind 2 query, $1 \le n \le 10^9$.

For Kind 3 or Kind 4 query, $1 \le n \le 10^3$ and $n \le p$.

For all queries, $2 \le p \le 10^4$ and p is a prime.

Output

For each query, output a line which contains only one integer.

Example

Input	Output
1 2 2 2 2 3 3 2 5 4 2 3	1 2 1 1

Problem AK. Help Yourself

Input File	help.in
Output File	help.out

Bessie has been given N ($1 \leq N \leq 10^5$) segments on a 1D number line. The ith segment contains all reals x such that $l_i \leq x \leq r_i$.

Define the **union** of a set of segments to be the set of all x that are contained within at least one segment. Define the **complexity** of a set of segments to be the number of connected regions represented in its union, raised to the power of K ($2 \leq K \leq 10$).

Bessie wants to compute the sum of the complexities over all 2^N subsets of the given set of N segments, modulo $10^9 + 7$.

Normally, your job is to help Bessie. But this time, you are Bessie, and there is no one to help you. Help yourself!

SCORING

- Test case 2 satisfies $N \leq 16$.
- Test cases 3-5 satisfy $N \leq 1000$, $K = 2$.
- Test cases 6-8 satisfy $N \leq 1000$.
- For each $T \in [9, 16]$, test case T satisfies $K = 3 + (T - 9)$.

INPUT FORMAT (file help.in):

The first line contains N and K.

Each of the next N lines contains two integers l_i and r_i. It is guaranteed that $l_i < r_i$ and all l_i, r_i are distinct integers in the range $1 \ldots 2N$.

OUTPUT FORMAT (file help.out):

Output the answer, modulo $10^9 + 7$.

Input	Output
3 2 1 6 2 3 4 5	10

The complexity of each nonempty subset is written below.

$$\{[1,6]\} \implies 1, \{[2,3]\} \implies 1, \{[4,5]\} \implies 1$$

$$\{[1,6],[2,3]\} \implies 1, \{[1,6],[4,5]\} \implies 1, \{[2,3],[4,5]\} \implies 4$$

$$\{[1,6],[2,3],[4,5]\} \implies 1$$

The answer is $1+1+1+1+1+4+1=10$.

Problem credits: Benjamin Qi

Problem AL. 黑白棋

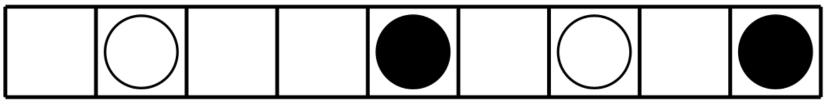
Time Limit 1000 ms
Mem Limit 128000 kB

Description

Little A and Little B have cooked up a brand-new game.

This game takes place on a $1 \times n$-sized board, with k pieces lined up—half are black, half are white.

The leftmost pieces are white, the rightmost are black, and neighboring pieces always alternate colors.



Little A moves the white pieces, Little B moves the black ones. White pieces can only move right, black pieces only move left. Each turn, a player must move between 1 and d pieces.

When moving a piece, it can't jump over the pieces on either end, and it can't step off the board. If a player can't make a move on their turn, they lose.

They take turns moving, with Little A going first. How many initial piece arrangements guarantee a win for Little A?

Input

Three numbers n, k, d are given as input.

Output

Output the total number of initial setups where Little A wins. The answer should be given modulo $10^9 + 7$.

Sample 1

Input	Output
10 4 2	182

Hint

- For 30% data, there are $k = 2$.
- For 100% data, there are $1 \le d \le k \le n \le 10^4$; k is even, $k \le 100$.

Problem AM. 移动金币

Time Limit 1000 ms
Mem Limit 512000 kB

Description

Alice and Bob are about to play a game. They take turns, with Alice going first.
On a player's turn, they must pick one coin and slide it any number of spaces to the left—at least one space.
Coins can't jump over other coins or move off the board.

The board has $1 \times n$ squares and initially holds m coins, each occupying a unique square (no stacking allowed).

If a player faces a turn where no moves are possible (which means the m coins are all packed into the leftmost m squares), that player loses.
Both Alice and Bob are super smart and always make the best moves possible. Given this, how many starting setups guarantee that Alice will win?

Input

One line containing two positive integers: n and m, as described above.

Output

Output a single integer — the count of initial configurations where Alice, moving first, can force a win. Since the number might be huge, output the result modulo $10^9 + 9$.

Sample 1

Input	Output
`10 3`	`100`

Sample 2

Input	Output
199 43	981535230

Sample 3

Input	Output
99999 47	39178973

Hint

Subtask 1: (50 points) $1 \le n \le 250$ and $1 \le m \le 50$.

Subtask 2: (50 points) $1 \le n \le 150000$ and $1 \le m \le 50$.

Problem AN. 强迫症

Time Limit 1000 ms
Mem Limit 524288 kB

Background

Little L is a hardcore perfectionist.

Because of his extreme OCD, whenever he draws, he insists on placing points perfectly on a circle.

Description

One day, he asked Little H and Little W this question:

If there are n distinct points on a circle, labeled in order from 1 to n, how many ways can you connect them to form a tree?

Little H & Little W: Isn't this a dumb question?

Little L: What if the edges are not allowed to cross?

Little H & Little W: Still a dumb question, right?

Little L: What if we swap "tree" for "graph"?

Little H & Little W: Still dumb, right?

Little L: Then what if each point has a weight a_i, and the weight of an edge connecting points (i, j) is $a_i \times a_j$? Find the **expected sum of all edge weights** of graphs satisfying the above conditions.

Little H & Little W: Still dumb, right?

After seeing his tough problem get crushed by the pros in seconds, Little L was pretty bummed out. To cheer him up, you need to solve this problem.

Note:

1. Two edges meeting at endpoints **do not count as crossing**.
2. A graph with **no edges at all (i.e., just n isolated points)** is also valid.
3. Points are numbered **clockwise from 1 to n**.

4. The graph **cannot have self-loops or multiple edges**.

Input

The first line contains a positive integer n, as described above.

The next line contains n non-negative integers, where the i-th number is a_i, representing the weight of the i-th point.

Output

A positive integer representing the result. The answer should be given modulo 998244353.

Sample 1

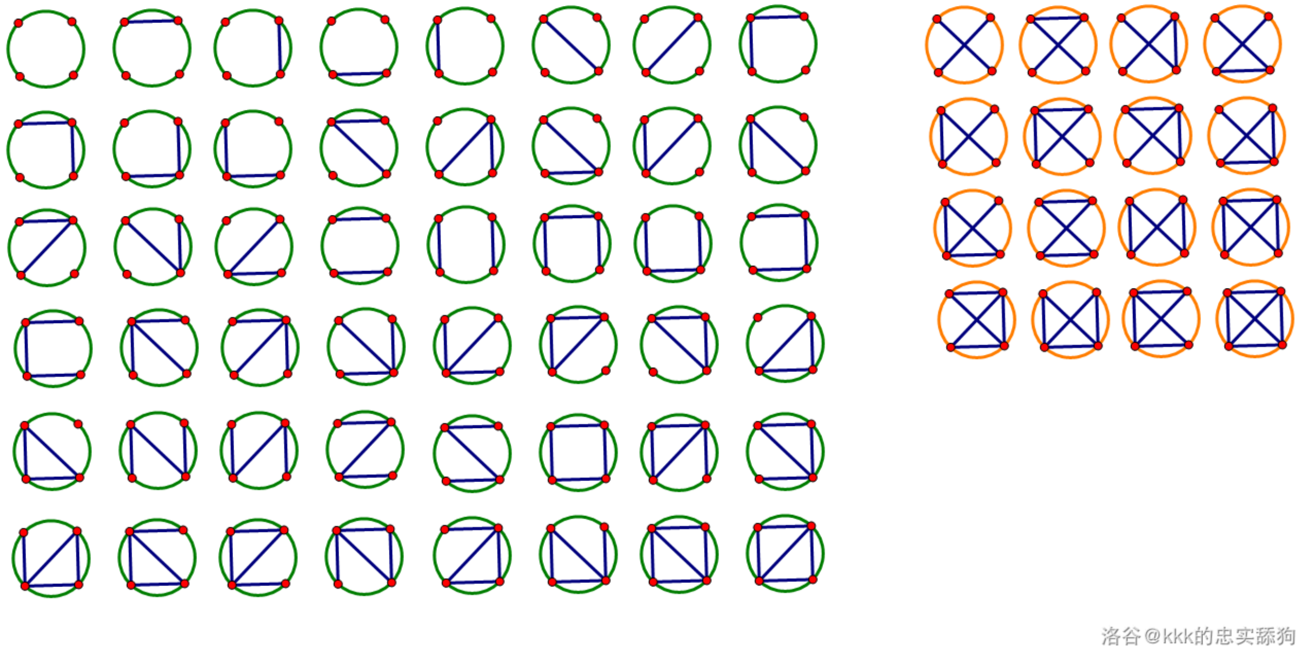
Input	Output
4 1 1 1 1	665496238

Sample 2

Input	Output
13 1 1 4 5 1 4 1 9 1 9 8 1 0	748867567

Hint

For Sample 1, all 64 graphs look like this:



Among them, the 48 graphs on the left are valid, while the 16 graphs on the right are invalid. All edges have weight 1.

The expected sum of edge weights is $\frac{8}{3}$. Taking modulo 998244353, the result is 665496238.

Data Constraints

This problem uses bundled test cases.

- Subtask 1 (10%): $n \le 6$.
- Subtask 2 (30%): $n \le 3000$.
- Subtask 3 (60%): No special restrictions.

For data where 100%, $2 \le n \le 10^5, 0 \le a_i \le 10^6$.

Time limits: Subtasks 1 and 2 have $1s$, Subtask 3 has $2s$.

If you don't know how to take modulo of a rational number, just Google "modular multiplicative inverse."

Problem AO. Train Tracking 2

Input File tracking2.in
Output File tracking2.out

Every day the express train goes past the farm. It has N carriages ($1 \le N \le 10^5$), each with a positive integer label between 1 and 10^9; different carriages may have the same label.

Usually, Bessie watches the train go by, tracking the carriage labels. But today is too foggy, and Bessie can't see any of the labels! Luckily, she has acquired the sliding window minimums of the sequence of carriage labels, from a reputable source in the city. In particular, she has a positive integer K, and $N - K + 1$ positive integers $c_1, \ldots, c_{N+1-K}$, where c_i is the minimum label among carriages $i, i+1, \ldots, i+K-1$.

Help Bessie figure out the number of ways to assign a label to each carriage, consistent with the sliding window minimums. Since this number may be very large, Bessie will be satisfied if you find its remainder modulo $10^9 + 7$.

Bessie's information is completely reliable; that is, it is guaranteed that there is at least one consistent way to assign labels.

INPUT FORMAT (file tracking2.in):

The first line consists of two space-separated integers, N and K. The subsequent lines contain the sliding window minimums $c_1, \ldots, c_{N+1-K}$, one per line.

OUTPUT FORMAT (file tracking2.out):

A single integer: the number of ways, modulo $10^9 + 7$, to assign a positive integer not exceeding 10^9 to each carriage, such that the minimum label among carriages $i, i+1, \ldots, i+K-1$ is c_i for each $1 \le i \le N - K + 1$.

Input	Output
4 2 999999998 999999999 999999998	3

Problem credits: Dhruv Rohatgi

Problem AP. 数列

Time Limit 1000 ms
Mem Limit 128000 kB

Description

Little T has recently started dabbling in stock trading and got a hot tip: F company's stock is about to skyrocket. The stock price each day is a positive integer and, due to some real-world constraints, can never exceed N. Over the K days of this wild surge, Little T noticed that except for the first day, each day's price is higher than the previous day's, but the increase (the difference between today's and yesterday's price) never goes beyond M. Here, M is a positive integer, and these parameters satisfy $M(K-1) < N$.
Unfortunately, Little T forgot the exact prices for these K days, and now he wants to figure out how many possible price sequences there could be over these K days.

Input

Just one line with four numbers separated by spaces: N, K, M, P. For details about P, check the explanation of P in the output format below. The input guarantees that the data for 20% meets $M, N, K, P \leq 20000$, and the data for 100% meets $M, K, P \leq 10^9$, $N \leq 10^{18}$.

Output

Just a single number representing the count of possible stock price sequences over these K days, modulo P.

Sample 1

Input	Output
7 3 2 997	16

Hint

Sample Explanation

The output 16 in the sample means there are 16 possible stock price sequences for the given input:

$\{1, 2, 3\}$, $\{1, 2, 4\}$, $\{1, 3, 4\}$, $\{1, 3, 5\}$, $\{2, 3, 4\}$, $\{2, 3, 5\}$, $\{2, 4, 5\}$, $\{2, 4, 6\}$, $\{3, 4, 5\}$, $\{3, 4, 6\}$, $\{3, 5, 6\}$, $\{3, 5, 7\}$, $\{4, 5, 6\}$, $\{4, 5, 7\}$, $\{4, 6, 7\}$, $\{5, 6, 7\}$.

Problem AQ. LCS Again

Time Limit 2000 ms
Mem Limit 262144 kB

You are given a string S of length n with each character being one of the first m lowercase English letters.

Calculate how many different strings T of length n composed from the first m lowercase English letters exist such that the length of LCS (longest common subsequence) between S and T is $n - 1$.

Recall that LCS of two strings S and T is the longest string C such that C both in S and T as a subsequence.

Input

The first line contains two numbers n and m denoting the length of string S and number of first English lowercase characters forming the character set for strings ($1 \le n \le 100\,000, 2 \le m \le 26$).

The second line contains string S.

Output

Print the only line containing the answer.

Examples

Input	Output
3 3 aaa	6

Input	Output
3 3 aab	11

Input	Output
1 2 a	1

Input	Output
10 9 abacadefgh	789

Note

For the first sample, the 6 possible strings T are: `aab`, `aac`, `aba`, `aca`, `baa`, `caa`.

For the second sample, the 11 possible strings T are: `aaa`, `aac`, `aba`, `abb`, `abc`, `aca`, `acb`, `baa`, `bab`, `caa`, `cab`.

For the third sample, the only possible string T is `b`.

Problem AR. 你的生命已如风中残烛

Time Limit 2000 ms

Mem Limit 262144 kB

Description

Midterms are over, and Rikka, who's no longer scared of anything, decided to skip math today and instead challenged Yuta to a duel of Yu-Gi-Oh!

"You're empty-handed, no cards on the field, and only 100 life points left. What can you possibly do now?"

"Summon the Exodia!"

"Huh?"

But Rikka's deck is just too weak. After some analysis, she realized that in a single turn, her deck can't pull off an OTK (One Turn Kill) unless she's got the protagonist's luck:

The Sealed Exodia

— If you have all five pieces — "Sealed Right Leg," "Sealed Left Leg," "Sealed Right Arm," "Sealed Left Arm," and this card — in your hand, you win the duel instantly.

However, since Rikka isn't a fancy whale who spends tons of money, she's certain that one of these five cards is buried at the very bottom of her deck. So now she faces a tough challenge: she needs to draw her entire deck in a single turn.

Her deck has $m + 1$ cards in total, but since the last card is fixed, we only consider the first m cards.

Among these m cards, there are n special cards and $m - n$ normal cards. Each special card has an attribute w_i that tells how many extra cards you can draw after playing it. Luckily, Rikka found that these cards satisfy $\sum_{i=1}^{n} w_i = m$, so she can safely draw cards without worrying about running out of cards.

Because these m cards are shuffled, there are $m!$ different possible deck orders.

Now the turn begins: Rikka draws one card from the deck, then keeps playing cards from her hand. If she plays a special card, she draws more cards accordingly, continuing until she draws the last card and wins, or runs out of cards in her hand and loses.

For example, if the deck is $\{4, 0, 0, 2, 0, 0, 0\}$ (where 0 represents a normal card, other numbers represent special cards, and the last 0 is the last Exodia piece), the play might go like this:

- Draw one card, hand is $\{4\}$.
- Play $\{4\}$, draw four cards, hand is $\{0, 0, 2, 0\}$.
- Play $\{2\}$, draw two more cards, and since the last Exodia piece is drawn, Rikka wins.
- But if the deck is $\{2, 0, 0, 4, 0, 0, 0\}$, it's easy to see Yuta wins instead.

Now, Rikka wants to know, out of these $m!$ possible deck orders, how many will let her win.

Input

The first line contains an integer n.

The second line contains n positive integers separated by spaces: w_i.

You can use this input to calculate $m = \sum w_i$ yourself.

Output

Output a single integer representing the answer. Since the answer might be huge, output it modulo 998 244 353.

Sample 1

Input	Output
1 5	24

Sample 2

Input	Output
6 1 2 3 4 5 6	90337303

Hint

Explanation for Sample 1

Among all $m!$ possible deck orders, only 24 decks of the form $\{5, 0, 0, 0, 0, 0\}$ satisfy the winning condition.

Data Constraints

For 10% data, $m \leq 10$.

For 30% data, $n \leq 8$.

For 50% data, $n \leq 15$.

For 70% data, $n \leq 30$.

For 100% data, $n \leq 40$, $1 \leq w_i \leq 10^5$. Also guaranteed is $m - n \geq 4$ because the deck must contain these 5 Exodia pieces.

Problem AS. Cards

Time Limit 4000 ms
Mem Limit 262144 kB

Consider the following experiment. You have a deck of m cards, and exactly one card is a joker. n times, you do the following: shuffle the deck, take the top card of the deck, look at it and return it into the deck.

Let x be the number of times you have taken the joker out of the deck during this experiment. Assuming that every time you shuffle the deck, all $m!$ possible permutations of cards are equiprobable, what is the expected value of x^k? Print the answer modulo 998244353.

Input

The only line contains three integers n, m and k ($1 \le n, m < 998244353$, $1 \le k \le 5000$).

Output

Print one integer — the expected value of x^k, taken modulo 998244353 (the answer can always be represented as an irreducible fraction $\frac{a}{b}$, where $b \mod 998244353 \neq 0$; you have to print $a \cdot b^{-1} \mod 998244353$).

Examples

Input	Output
1 1 1	1

Input	Output
1 1 5000	1

Input	Output
2 2 2	499122178

Input	Output
998244352 1337 5000	326459680

Problem AT. 魔力环

Time Limit 1000 ms
Mem Limit 512000 kB

Background

wkr is an OI enthusiast from the magical city who loves collecting black and white magic beads.

Description

wkr dreams of creating a ring made up of n magic beads. But he's not interested in just any ordinary ring—he has some very specific demands:

- The ring must contain **exactly** m black magic beads and $n - m$ white magic beads.
- Since wkr thinks black beads shouldn't cluster too tightly, he insists that the ring **cannot** have any run of consecutive black beads longer than k.

Only a ring that meets these conditions is truly enchanting in wkr's eyes.

But there might be more than one way to arrange such a ring. wkr is curious: how many distinct rings satisfy his criteria? However, he's not a fan of doing the math himself, so he's counting on your cleverness to find the answer.

Here, two rings are considered different if you cannot get one from the other by simply rotating it.

Because the answer could be huge, please output the count modulo $998,244,353$.

Input

The input consists of 1 lines containing 3 non-negative integers n, m, k in total, representing the parameters described above. Adjacent numbers are separated by a single space.

Output

Output 1 lines with 1 integers, each representing the number of rings that meet the requirements.

Sample 1

Input	Output
6 3 2	3

Sample 2

Input	Output
17 8 6	1421

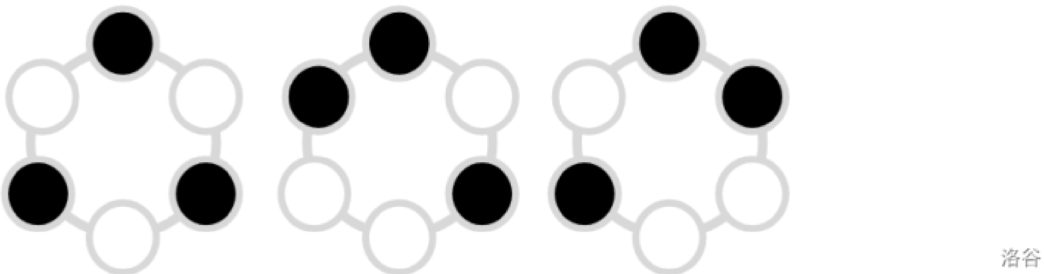
Sample 3

Input	Output
50000 20000 1	683811528

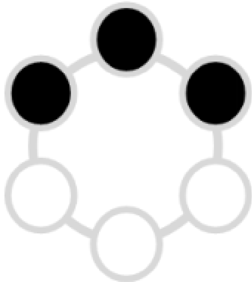
Hint

Explanation for Sample 1

We have a ring made of 6 magic beads, with exactly 3 black beads and 3 white beads, and no run of black beads longer than 2. There are 3 distinct rings that satisfy these conditions, illustrated in the figure below.



The ring shown below does not meet wkr's requirements because it contains a run of black beads longer than 2.



Subtasks

All test cases satisfy $1 \le n, k \le 10^5, 0 \le m \le 10^5$ and $m \le n$.

This problem uses bundled testing, with 7 subtasks. The points and constraints for each subtask are as follows:

- Subtask 1 (3 points): $m = 0$.
- Subtask 2 (5 points): $n \le 4$.
- Subtask 3 (8 points): $n \le 18$.
- Subtask 4 (7 points): $m = 2$.
- Subtask 5 (19 points): $k = 1$.
- Subtask 6 (27 points): $\gcd(n, m) \le 2$.
- Subtask 7 (31 points): No special restrictions.

Source

MtOI2018 Lost House Water-themed Contest T6

Problem setter: Imagine

72679

Problem AU. 十二重计数法

Time Limit 2000 ms
Mem Limit 262144 kB

Background

Combinatorics is an ancient and fascinating branch of mathematics.

Legend has it that 114514 years ago, a deity named Yi Ai descended to Earth and discovered humans—another intelligent species.

Finding this quite intriguing, she decided to speed up human civilization by introducing a class of counting problems known as the Twelvefold Counting—marking the dawn of combinatorics.

Mastering these problems is the key to diving deeper into the world of combinatorics.

Description

Given n balls and m boxes, all balls need to be placed into the boxes.
There are some constraints—how many ways can you arrange the balls? (Order of placement doesn't matter)

Here are the constraints:

I: Balls are all distinct, boxes are all distinct.
II: Balls are all distinct, boxes are all distinct, with at most one ball per box.
III: Balls are all distinct, boxes are all distinct, with at least one ball in each box.

IV: Balls are all distinct, boxes are identical.
V: Balls are all distinct, boxes are identical, with at most one ball per box.
VI: Balls are all distinct, boxes are identical, with at least one ball in each box.

VII: Balls are identical, boxes are all distinct.
VIII: Balls are identical, boxes are all distinct, with at most one ball per box.
IX: Balls are identical, boxes are all distinct, with at least one ball in each box.

X: Balls are identical, boxes are identical.
XI: Balls are identical, boxes are identical, with at most one ball per box.
XII: Balls are identical, boxes are identical, with at least one ball in each box.

Since the answer can be huge, output the result modulo 998244353.

Input

A single line containing two positive integers n, m

Output

Print twelve lines, each containing one integer—the answer corresponding to each constraint above.

Sample 1

Input	Output
13 6	83517427 0 721878522 19628064 0 9321312 8568 0 792 71 0 14

Hint

【Data Range】
For 100% data, $1 \leq n, m \leq 2 \times 10^5$.

orz EntropyIncreaser

Problem AV. 有标号 DAG 计数

Time Limit 1000 ms
Mem Limit 128000 kB

Description

Count the number of directed acyclic graphs (DAGs) with n labeled nodes, where the graph is weakly connected (meaning if you replace all directed edges with undirected ones, the resulting graph is connected). Output the result modulo 998244353.

Input

A single integer T.

Output

A total of T lines, where the answer for $n = i$ is printed on the $i (1 \le i \le T)$-th line.

Sample 1

Input	Output
5	1 2 18 446 26430

Hint

The first point $T = 2000$.

The second point $T = 100000$.

Problem AW. Bandit Blues

Time Limit 3500 ms
Mem Limit 262144 kB

Japate, while traveling through the forest of Mala, saw N bags of gold lying in a row. Each bag has some **distinct** weight of gold between 1 to N. Japate can carry only one bag of gold with him, so he uses the following strategy to choose a bag.

Initially, he starts with an empty bag (zero weight). He considers the bags in some order. If the current bag has a higher weight than the bag in his hand, he picks the current bag.

Japate put the bags in some order. Japate realizes that he will pick A bags, if he starts picking bags from the front, and will pick B bags, if he starts picking bags from the back. By picking we mean replacing the bag in his hand with the current one.

Now he wonders how many permutations of bags are possible, in which he picks A bags from the front and B bags from back using the above strategy.

Since the answer can be very large, output it modulo 998244353.

Input

The only line of input contains three space separated integers N ($1 \le N \le 10^5$), A and B ($0 \le A, B \le N$).

Output

Output a single integer — the number of valid permutations modulo 998244353.

Examples

Input	Output
1 1 1	1

Input	Output
2 1 1	0

Input	Output
2 2 1	1

Input	Output
5 2 2	22

Note

In sample case 1, the only possible permutation is $[1]$

In sample cases 2 and 3, only two permutations of size 2 are possible: $\{[1, 2], [2, 1]\}$. The values of a and b for first permutation is 2 and 1, and for the second permutation these values are 1 and 2.

In sample case 4, out of 120 permutations of $[1, 2, 3, 4, 5]$ possible, only 22 satisfy the given constraints of a and b.

Problem AX. 命运

Time Limit 2000 ms
Mem Limit 1048576 kB

Description

Heads up: At the end of this description, you'll find a neat, formal summary of the problem.

In a far-flung future, physicists finally cracked the natural laws of time and causality. They discovered that even before someone is born, we can theoretically peek into their life story. In other words, physics lets us "predict" a person's fate to some extent.

Picture a person's fate as a rooted tree made up of moments in time T: the root node marks birth, and the leaves mark death. Every non-leaf node u sprouts one or more children $v_1, v_2, \ldots, v_{c_u}$, representing the different choices c_u a person can make at the moment u it stands for. Formally, a choice is an edge (u, v_i) in the tree, where u is the parent of v_i.

A person's life is a path from birth (the root) to death (a leaf) without revisiting any node. Any subpath along this journey that includes at least one edge is a slice of their **life experience**. The entire collection of all possible life experiences they could have—walking every possible path—is called the set of **potential life experiences**. Put simply, all potential life experiences are paths from u to v where $u, v \in T$, $u \neq v$, and u is an ancestor of v. Mathematically, such a potential life experience is denoted as an ordered pair (u, v); the set of all potential life experiences in the tree T is called $\mathcal{P}_T$.

Physics doesn't just let us observe this fate tree; it also lets us analyze which potential life experiences are "important." Every choice a person makes—that is, every edge in the tree—can be either **important** or **not important**. A potential life experience is deemed important if its path contains at least one important edge. We can observe some potential life experiences that are important: in other words, we have a set $\mathcal{Q} \subseteq \mathcal{P}_T$ where every potential life experience $(u, v) \in \mathcal{Q}$ in it is important.

The shape of the tree T is already computed, and the set $\mathcal{Q}$ is observed. The uncertainty about a person's fate has shrunk dramatically, but it's still huge—so let's crunch the numbers! Given the tree T and the set $\mathcal{Q}$, how many different ways can we mark each edge as important or not important so that all observed constraints from $\mathcal{Q}$ are satisfied? That is, for every $(u, v) \in \mathcal{Q}$, there exists at least one edge on the path from u to v marked as important.

Formally: Given a tree $T = (V, E)$ and a set of node pairs $\mathcal{Q} \subseteq V \times V$, where for every $(u, v) \in \mathcal{Q}$ we have $u \neq v$ and u is an ancestor of v in T. Let V and E be the node set and edge set of T respectively. Find the number of functions $f\colon E \to \{0, 1\}$ (assigning each edge $e \in E$ a value of either 0 or 1) such that for every $(u, v) \in \mathcal{Q}$, there exists an edge e on the path from u to v with $f(e) = 1$. Since the answer can be huge, output it modulo $998,244,353$ (a prime number).

Input

Read input from standard input.

The first line contains a positive integer n, the size of the tree T. The nodes are numbered from 1 to n, with node 1 as the root.

The next $n - 1$ lines each contain two space-separated integers x_i, y_i, representing an edge between nodes x_i and y_i. Note that the edges are undirected and their direction is not guaranteed.

Then comes a line with a non-negative integer m, indicating how many observed pieces of information we have.

The next m lines each contain two space-separated integers u_i, v_i, representing $(u_i, v_i) \in \mathcal{Q}$. **Heads up**: the input may contain duplicate information, meaning there might be pairs $i \neq j$ where $u_i = u_j$ and $v_i = v_j$.

Refer to the constraints table at the end of the problem for input size and limits.

Output

Output to standard output.

Print a single integer: the number of valid assignments modulo $998,244,353$.

Sample 1

Input	Output
5 1 2 2 3 3 4 3 5 2 1 3 2 5	10

Sample 2

Input	Output
15 2 1 3 1 4 3 5 2 6 3 7 6 8 4 9 5 10 7 11 5 12 10 13 3 14 9 15 8 6 3 12 5 11 2 5 3 13 8 15 1 13	960

Hint

Explanation for Sample 1

There are 16 valid assignments in total. The following 6 assignments fail to meet the requirements:

- Edge $(1, 2)$, $(2, 3)$, $(3, 5)$ marked as not important, edge $(3, 4)$ marked as important: no constraint from set $\mathcal{Q}$ is satisfied.
- Edge $(1, 2)$, $(2, 3)$, $(3, 4)$, $(3, 5)$ marked as not important: no constraint from set $\mathcal{Q}$ is satisfied.
- Edge $(1, 2)$, $(2, 3)$ marked as not important, edge $(3, 4)$, $(3, 5)$ marked as important: constraint $(1, 3)$ in set $\mathcal{Q}$ is not satisfied.

- Edge $(1,2)$, $(2,3)$, $(3,4)$ marked as not important, edge $(3,5)$ marked as important: constraint $(1,3)$ in set Q is not satisfied.
- Edge $(2,3)$, $(3,5)$ marked as not important, edge $(1,2)$, $(3,4)$ marked as important: constraint $(2,5)$ in set Q is not satisfied.
- Edge $(2,3)$, $(3,4)$, $(3,5)$ marked as not important, edge $(1,2)$ marked as important: constraint $(2,5)$ in set Q is not satisfied.
- Under all other assignments, all constraints in set Q are satisfied.

Sample 3

See the contestant directory files destiny/destiny3.in and destiny/destiny3.ans.

Sample 4

See the contestant directory files destiny/destiny4.in and destiny/destiny4.ans.

Test ID	n	m	Is T a complete binary tree?
$1 \sim 4$	≤ 10	≤ 10	No
5	≤ 500	≤ 15	No
6	$\le 10^4$	≤ 10	No
7	$\le 10^5$	≤ 16	No
8	$\le 5 \times 10^5$	≤ 16	No
9	$\le 10^5$	≤ 22	No
10	$\le 5 \times 10^5$	≤ 22	No
11	≤ 600	≤ 600	No
12	$\le 10^3$	$\le 10^3$	No
$13 \sim 14$	$\le 2 \times 10^3$	$\le 5 \times 10^5$	No
$15 \sim 16$	$\le 5 \times 10^5$	$\le 2 \times 10^3$	No
$17 \sim 18$	$\le 10^5$	$\le 10^5$	Yes
19	$\le 5 \times 10^4$	$\le 10^5$	No
20	$\le 8 \times 10^4$	$\le 10^5$	No
$21 \sim 22$	$\le 10^5$	$\le 5 \times 10^5$	No

Test ID	n	m	Is T a complete binary tree?
$23 \sim 25$	$\le 5 \times 10^5$	$\le 5 \times 10^5$	No

Test Constraints

All data satisfy: $n \le 5 \times 10^5$, $m \le 5 \times 10^5$. The input forms a tree, and for every $1 \le i \le m$, u_i is always an ancestor of v_i.

Complete Binary Tree: In this problem, a full binary tree is one where every non-leaf node has both left and right children, and all leaves are at the same depth. When nodes in a full binary tree are numbered top-down, left-to-right, the subtree formed by the smallest numbered nodes is called a complete binary tree.

Problem AY. 抛硬币

Time Limit 1000 ms
Mem Limit 128000 kB

Description

Little A and Little B are best buddies who love to hang out together. Recently, Little B got hooked on a gacha mobile game, grinding levels every day and totally neglecting his studies. After months of playing, he hasn't pulled a single SSR, which has him seriously questioning his luck. Diligent Little A wants to help Little B quit the game and focus on studying, so he comes up with a plan: they'll both toss coins k times, and if Little A gets more heads than Little B, he wins.

But here's the twist—Little A used to be addicted to gacha games too and never got a UR, so he's not exactly lucky either. To tip the scales, he decides to cheat by secretly tossing a few extra coins when Little B isn't looking. Of course, he won't toss too many to avoid suspicion. Now, Little A wants to know in how many possible scenarios he can beat Little B. Since the number might be huge, just output the last k digits of the answer in decimal.

Input

Multiple test cases. For each test case, input three integers a, b, k representing the number of coin tosses for Little A, the number of coin tosses for Little B, and how many digits of the final answer to keep.

Output

For each test case, output a number representing the last k digits of the total number of winning scenarios for Little A. If the number has fewer than k digits, pad it with leading 0 s.

Sample 1

Input	Output
2 1 9 3 2 1	000000004 6

Hint

For the first test case, when Little A tosses 2 coins and Little B tosses 1 coins, there are 4 ways for Little A to get more heads than Little B.

$(01, 0), (10, 0), (11, 0), (11, 1)$

For the second test case, when Little A tosses 3 coins and Little B tosses 2 coins, there are 16 ways for Little A to get more heads than Little B.

$(001, 00), (010, 00), (100, 00), (011, 00), (101, 00), (110, 00), (111, 00), (011, 01)$

$(101, 01), (110, 01), (111, 01), (011, 10), (101, 10), (110, 10), (111, 10), (111, 11)$

Constraints

10% data satisfy $a, b \le 20$.

30% data satisfy $a, b \le 100$.

70% data satisfy $a, b \le 10^5$, among which 20% data satisfy $a = b$.

100% data satisfy $1 \le a, b \le 10^{15}$, $b \le a \le b + 10^4$, $1 \le k \le 9$, with the number of test cases not exceeding 10.